Software Requirements Specification

for

Dataset Integrity Analyzer

An AI-Based Framework for Dataset Health Assessment and Fairness Evaluation

Version 1.0 approved

Prepared by Srijan Bajpai, 24BRS1212

Organization: VIT Chennai, Chennai

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5. Other Nonfunctional Requirements

○ 5.1 Performance Requirements

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Revision History

Name	Date	Reason For Changes	Version
Srijan Bajpai	14-07-2026	Initial baseline SRS for Software Engineering Lab project	1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for **Dataset Integrity Analyzer (DIA), Version 1.0** — an AI-based framework that automatically assesses the health, quality, fairness and training-readiness of image classification datasets.

This document covers the complete Version 1.0 system, which comprises: the dataset ingestion subsystem, the analysis engine (fairness, class balance, image quality, duplication, diversity, label consistency), the Bias Impact Prediction module, the AI Readiness Score engine, the recommendation engine, and the reporting/visualisation front-end. It does **not** cover automated dataset *repair* (e.g. synthetic image generation or automated re-labelling), which is explicitly deferred to a future release.

The SRS serves as the agreed baseline between the developer, the course faculty/reviewer, and any downstream evaluator, and is the reference against which the system will be verified.

1.2 Document Conventions

- Requirements are uniquely tagged as **REQ-\<n\>** and are numbered sequentially within each system feature (Section 4).
- Each system feature carries a priority of **High**, **Medium**, or **Low**. Priority is inherited by all requirements under that feature *unless* a requirement explicitly overrides it.
- The keywords **shall** (mandatory), **should** (recommended), and **may** (optional) are used in accordance with IEEE Std 830-1998.
- Items not yet decided are marked **TBD-\<n\>** and are collected in Appendix C.

- Monospace text denotes file paths, identifiers, formats, or code entities.

1.3 Intended Audience and Reading Suggestions

Reader	Suggested Reading Path
Course faculty / project reviewer	Sections 1, 2, then 4 for scope of delivery; Section 5.4 for quality targets.
Developer (implementer)	Full document, sequentially. Sections 3 and 4 are the primary implementation contract.
Tester / evaluator	Section 4 (functional requirements as test conditions) and Section 5 (measurable NFR thresholds).
ML researcher / end user	Sections 1.4, 2.2, 2.3, and 4.11 (report contents).
Conference paper co-author	Sections 1.4, 4.8 (novelty), 1.5 (literature base), and Appendix B.

The document proceeds from context (Section 2) to interfaces (Section 3), to detailed behaviour (Section 4), to quality constraints (Section 5).

1.4 Product Scope

Modern machine learning systems inherit the deficiencies of the data on which they are trained. Publicly available and organisational image datasets frequently exhibit class imbalance, demographic skew, duplicate samples, degraded image quality, inconsistent labels, and insufficient visual diversity. These defects typically remain undetected until after model training, at which point they manifest as reduced accuracy, unfair sub-group performance, wasted computation, and untrustworthy deployment behaviour.

1.4.1 The Gap: Fragmentation of Dataset Quality Assessment

The existing landscape of dataset assessment is **fragmented along a single axis per tool**. Each mature tool or published method addresses exactly *one* dimension of dataset health, in isolation from the others:

Dimension	Representative existing tooling / method	What it does **not** cover
Fairness / bias	AIF360, Fairlearn, Aequitas, Gender Shades methodology	Image quality, duplication, diversity, label noise

Dimension	Representative existing tooling / method	What it does **not** cover
Class imbalance	`imbalanced-learn`, SMOTE and its variants	Fairness, quality, duplication, diversity
Duplicate detection	`imagededup`, FiftyOne uniqueness, perceptual-hash pipelines	Fairness, imbalance, quality, label noise
Image quality	BRISQUE, NIQE, PIQE, no-reference IQA literature	Fairness, imbalance, duplication, diversity
Label noise	Cleanlab / confident learning	Fairness, imbalance, quality, diversity
Documentation	Datasheets for Datasets, Model Cards	Provides a *disclosure template*, performs **no measurement at all**

The practical consequence for a researcher is that a genuinely rigorous pre-training audit today requires **assembling five or six separate tools**, each with its own data format, its own dependency stack, its own scale of output, and its own notion of severity. There is no shared scale on which "an Imbalance Ratio of 8.2," "a 4.1 % near-duplicate rate," and "a mean BRISQUE of 47" can be compared against one another; there is no way to know which of them to fix *first*; and there is no mechanism at all for detecting **cross-dimensional interactions** — for instance, that the low-quality images in a dataset are *concentrated in one demographic sub-group*, which is a fairness defect that no fairness tool can see (because it inspects only labels) and no quality tool can see (because it inspects only pixels). Such interactions are invisible to every single-axis tool by construction, and are precisely the defects most likely to produce a model that is accurate in aggregate and unfair in the particular.

Furthermore, most existing tooling is **model-coupled**: it diagnoses bias in a *trained classifier* rather than in the *dataset*, and therefore reports the problem only after the GPU budget has already been spent.

1.4.2 The Product: A Unified Dataset Health Framework

Dataset Integrity Analyzer (DIA) addresses this gap by **integrating dataset quality assessment into a single platform**. It is a *unified dataset health framework* that simultaneously evaluates **fairness, class balance, image quality, duplication, diversity, and label integrity**, reconciles them onto **one common 0–100 scale**, produces a single **overall dataset integrity score (the AI Readiness Score)**, and issues **prioritised, actionable recommendations to improve the dataset before it is used for AI model development.**

Given an image classification dataset, the system:

1. Profiles class distribution, inferred demographic attributes, and image-level quality characteristics.

2. Detects duplicates, near-duplicates, train–test leakage, low-diversity clusters, and label anomalies.

3. **Detects cross-dimensional confounds** that no single-axis tool can observe — quality × demographic, class × attribute, duplication × split, diversity × class.

4. Computes a composite, interpretable **AI Readiness Score (0–100)** with six transparent, auditable sub-scores.

5. **Predicts the expected degradation in downstream model accuracy and sub-group fairness** attributable to the measured defects.

6. Emits a **Dataset Health Report** with prioritised, cost-aware corrective actions.

1.4.3 Novel Contributions

The project makes **two** distinct contributions, of which the first is architectural and the second is methodological.

- **C1 — Unification (breadth).** DIA is, to the best of the authors' knowledge, the first system to assess *all six* dimensions of dataset health **within one pipeline, over one shared representation, on one comparable scale, yielding one overall dataset integrity score.** The contribution is not merely convenience-of-packaging. Unification is what makes three things *possible* that are unavailable to any composition of single-axis tools: (a) **cross-dimensional confound detection** (§4.3, §4.4, §4.6), which requires the fairness and quality analyses to share a sample index; (b) **commensurable prioritisation** — a principled answer to "which defect do I fix first?" — which requires all defects to be expressed on one scale (§4.9, §4.10); and (c) the bias descriptor vector of C2, which is only constructible because all six dimensions have been measured over the same samples (§4.8). Each of these is unobtainable by running five tools side by side.
- **C2 — Impact prediction (depth).** Rather than merely *reporting* bias, DIA **predicts its downstream consequence**: the expected drop in test accuracy and the expected worst-sub-group performance gap that a model trained on this dataset would exhibit. This converts an abstract statistic ("gender is 82:18") into an actionable engineering quantity ("expect ≈ 11 pp lower accuracy on the female sub-group"). The predictor is trained on a corpus of synthetically perturbed datasets with known injected defects and validated leave-one-dataset-out (§4.8).

C1 and C2 are mutually reinforcing rather than independent: C2 is **only feasible because of C1**, since the bias descriptor vector that drives the prediction is assembled from all six dimensions simultaneously. A tool that measured only fairness could not predict impact, because impact is a joint function of imbalance, quality, redundancy, diversity, and label noise together.

A third, secondary contribution (**C3**) is the reusable synthetic-perturbation benchmark harness built to train the predictor, released independently (§OR-9).

Objective. To design and develop an intelligent software system that automatically evaluates the integrity, quality, fairness, and readiness of machine learning datasets by applying computer vision, statistical analysis, and machine learning techniques within a single unified framework, thereby assisting researchers and developers in identifying dataset deficiencies before model training.

Scope boundary. Version 1.0 targets **image classification datasets** only (folder-per-class layouts and CSV/JSON manifest layouts). Object detection, segmentation, video, tabular, and NLP datasets are out of scope. The system *analyses* and *recommends*; it does not modify the user's dataset.

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9. C. Northcutt, L. Jiang, and I. Chuang, "Confident Learning: Estimating Uncertainty in Dataset Labels," *Journal of Artificial Intelligence Research*, vol. 70, pp. 1373–1411, 2021.

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12. P. Stock and M. Cisse, "ConvNets and ImageNet Beyond Accuracy: Understanding Mistakes and Uncovering Biases," *Proc. ECCV*, 2018.

13. J. Deng et al., "ImageNet: A Large-Scale Hierarchical Image Database," *Proc. IEEE CVPR*, 2009.

Single-axis dataset assessment tools — the fragmented baseline against which the unification contribution (C1) of this work is positioned (see §1.4.1):

14. R. K. E. Bellamy et al., "AI Fairness 360: An Extensible Toolkit for Detecting and Mitigating Algorithmic Bias," *IBM Journal of Research and Development*, vol. 63, no. 4/5, 2019. *(Fairness only.)*

15. S. Bird et al., "Fairlearn: A Toolkit for Assessing and Improving Fairness in AI," Microsoft Technical Report MSR-TR-2020-32, 2020. *(Fairness only.)*

16. P. Saleiro et al., "Aequitas: A Bias and Fairness Audit Toolkit," *arXiv:1811.05577*, 2018. *(Fairness only.)*

17. G. Lemaître, F. Nogueira, and C. K. Aridas, "Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning," *JMLR*, vol. 18, no. 17, pp. 1–5, 2017. *(Class imbalance only.)*

18. T. Jain, C. Lennan, Z. John, and D. Tran, "Imagededup," GitHub, 2019, https://github.com/idealo/imagededup. *(Duplication only.)*

19. B. E. Moore and J. J. Corso, "FiftyOne," GitHub, 2020, https://github.com/voxel51/fiftyone. *(Sample-level exploration and uniqueness; no unified integrity score, no impact prediction.)*

20. A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a 'Completely Blind' Image Quality Analyzer" (NIQE), *IEEE Signal Processing Letters*, vol. 20, no. 3, 2013. *(Image quality only.)*

21. N. Venkatanath, D. Praneeth, M. C. Bh, S. S. Channappayya, and S. S. Medasani, "Blind Image Quality Evaluation Using Perception Based Features" (PIQE), *Proc. NCC*, 2015. *(Image quality only.)*

22. M. Mitchell et al., "Model Cards for Model Reporting," *Proc. FAT**, 2019. *(Disclosure template; performs no measurement.)*

23. N. Sambasivan, S. Kapania, H. Highfill, D. Akrong, P. Paritosh, and L. M. Aroyo, "'Everyone wants to do the model work, not the data work': Data Cascades in High-Stakes AI," *Proc. ACM CHI*, 2021. *(Empirical motivation for pre-training dataset auditing.)*

Dataset sources:

24. IEEE DataPort, https://ieee-dataport.org — dataset repository used for evaluation datasets.

25. Kaggle Datasets, https://www.kaggle.com/datasets — source of UTKFace, CelebA, HAM10000, Chest X-Ray datasets.

1.6 Problem Analysis

This section presents a structured analysis of the problem the product addresses, prior to the statement of requirements in Sections 3–5. It follows the sequence: problem definition → root-cause analysis → stakeholder analysis → evidence of need → constraint analysis → feasibility → problem statement.

1.6.1 Problem Definition

The problem, stated precisely:

> Machine learning practitioners commit substantial time and compute to training models on datasets whose defects are unknown to them at the moment of commitment. Existing diagnostic tools each measure one dimension of dataset quality in isolation, on an incommensurable scale, and none predicts the downstream consequence of the defects they find. The practitioner therefore cannot answer three questions that must be answered *before* training begins: **(i) Is my dataset fit for purpose? (ii) If not, what is wrong with it? (iii) Of the things that are wrong, which should I fix first?**

The problem is **not** a lack of individual metrics — the literature supplies these in abundance (§1.5). The problem is the **absence of an integrated, decision-supporting layer** above them.

1.6.2 Root-Cause Analysis

Applying a *five-whys* decomposition to the observed symptom:

Level	Question	Answer
Symptom	—	Trained models exhibit poor accuracy, unfair sub-group performance, and inflated test scores that collapse in deployment.
Why 1	Why do models fail this way?	Because they inherit defects present in the training data — imbalance, demographic skew, duplicates, poor image quality, low diversity, label noise.
Why 2	Why are these defects present at training time?	Because they were never detected. The dataset was not audited before it was used.
Why 3	Why was it not audited?	Because a thorough audit currently requires assembling and reconciling five or six separate single-purpose tools, each with its own interface, dependency stack, and output format. The cost of the audit is perceived as exceeding its benefit.

Level	Question	Answer
Why 4	Why is the audit so costly?	Because no tool spans the dimensions. Each was built to answer one question well, and none was built to answer the practitioner's actual question — *"should I train on this?"*
Why 5 (root)	Why has no such tool been built?	Because the field's research incentives reward **novel metrics** (a new fairness measure, a better IQA score) over **integration** — and because integration is only valuable if the dimensions are measured over a *shared representation*, which is an architectural commitment no single-metric paper has cause to make.

Root cause: *Dataset quality assessment is fragmented by construction, and the fragmentation prevents both cross-dimensional reasoning and cost-effective auditing.*

This root cause maps directly onto the two contributions of this project: **C1 (unification)** attacks the fragmentation itself; **C2 (impact prediction)** attacks the cost–benefit calculation, by making the *consequence* of skipping the audit legible in advance.

1.6.3 Problem Decomposition

The root problem decomposes into six sub-problems, each of which is discharged by an identified system feature:

#	Sub-problem	Consequence if unaddressed	Discharged by
P1	Class and demographic representation are unmeasured	Model underperforms on minority classes and sub-groups	§4.3
P2	Image quality is unmeasured and may correlate with class or group	Model learns a quality artefact as a spurious shortcut	§4.4
P3	Duplicates and cross-split leakage are undetected	Test accuracy is inflated; the evaluation is invalid	§4.5
P4	Visual diversity and coverage are unmeasured	Model fails on conditions absent from training	§4.6
P5	Label errors are undetected	Model learns from a corrupted target signal	§4.7

#	Sub-problem	Consequence if unaddressed	Discharged by
P6	Defect *interactions* are invisible to all single-axis tools	The most consequential defects — those spanning two dimensions — are systematically missed	§4.8 (**C1**)
P7	The *downstream cost* of any defect is unknown	The practitioner cannot prioritise, and cannot justify the audit's cost	§4.9 (**C2**), §4.10, §4.11

P6 and P7 are the sub-problems no existing tool addresses, and they are therefore where this project's contribution lies. P1–P5 are individually solved in the literature; the project's task there is *integration*, not invention.

1.6.4 Stakeholder Analysis

Stakeholder	Interest in the problem	What they need from the solution	Influence
ML Researcher	Publishes results that must be reproducible and defensible against reviewer challenge on data quality	Quantitative, citable metrics; exportable artifacts; a datasheet-style summary	High
ML Practitioner / Engineer	Bears the compute cost of a failed training run	A pass/fail verdict and a prioritised fix list, delivered in minutes	High
Student / Learner	Learning to build models; unaware of data-quality failure modes	Plain-language explanations; guided defaults; teaching-by-diagnosis	Medium
Auditor / Compliance Reviewer	Must attest to fairness properties of a deployed system	A readable report they can consume without running the tool	Medium
Affected data subjects (indirect)	Are misclassified by models trained on data that under-represents them	*(Not a user.)* Their interest is served by the system existing at all, and is protected by REQ-15 and NFR-S3	None (voiceless)
Course faculty / reviewer	Assesses the project	Traceable requirements; demonstrable novelty; working artifact	High

The final row is deliberate. The people most harmed by biased datasets are the ones with **no channel to influence the tooling**, which is precisely why the ethical safeguards (§5.2) are stated as hard requirements rather than left to the developer's discretion.

1.6.5 Evidence of Need

The problem is not hypothetical. It is documented in the literature cited in §1.5:

- **Prevalence.** *Gender Shades* [4] demonstrated that commercial gender-classification systems exhibited error-rate disparities of up to 34 percentage points between demographic sub-groups — disparities traceable to the composition of the training data, not to the model architecture.
- **Undetected quality confounds.** Stock and Cisse [12] showed that models trained on ImageNet exploit incidental image properties as class shortcuts — the class of defect that §4.8 exists to detect and that no single-axis tool can see.
- **Systemic neglect.** Sambasivan et al. [23] documented, across 53 practitioner interviews, that data quality work is systematically under-prioritised relative to model work, and that the resulting *data cascades* surface late and expensively. This is direct empirical support for §1.6.2's Why-3 and Why-4.
- **Tooling gap.** The fragmentation table in §1.4.1 is itself the evidence for the gap: every mature tool in the space [14]–[21] addresses exactly one axis.
- **Recognised, but only as disclosure.** *Datasheets for Datasets* [3] and *Model Cards* [22] establish that the community accepts the need for dataset transparency — but both prescribe a **documentation template to be completed by hand**. Neither measures anything. They establish demand for the answer without supplying a means of computing it.

1.6.6 Constraint Analysis

Constraint	**Type**	**Impact on solution**
Single developer, no prior ML implementation experience	**Resource / skill**	Rules out from-scratch model training as the *core* of the system; favours composition of pretrained models and established libraries (§2.5, CON-2)
Single-semester schedule	**Schedule**	Forces incremental delivery; the system must be demonstrable before it is complete
Consumer-grade compute (single GPU or CPU-only)	**Resource**	Bounds C2's surrogate training corpus (CON-8); necessitates lightweight architectures

Constraint	Type	Impact on solution
Datasets may contain facial imagery	**Legal / ethical**	Mandates local-only processing (NFR-SEC1) and inference disclaimers (REQ-15)
Novelty required for publication	**Academic**	The system must do something no existing tool does — satisfied by C1 and C2
Requirements will change as capability is discovered	**Technical uncertainty**	See §1.7 — this is the dominant driver of process model selection

The final constraint is the critical one, and it motivates the next section.

1.6.7 Feasibility Assessment

Dimension	Assessment	Basis
Technical — P1–P5	**High.**	These reduce to statistics and calls to mature libraries (`pandas`, `OpenCV`, `imagehash`, `scikit-learn`, `cleanlab`) plus *inference* on pretrained models. No model training is required.
Technical — P6 (C1)	**High.**	Confound detection is a set of hypothesis tests over a joined table. The difficulty is *architectural* (maintaining a shared sample index), not algorithmic — and architecture is a design discipline, not an ML one.
Technical — P7 (C2)	**Medium-Low.**	Requires generating perturbed dataset variants, training a surrogate CNN on each, and fitting a regressor. This is the only component demanding genuine ML engineering, and it carries real risk of schedule overrun.
Schedule	**Conditional.**	Feasible **only** if C2 is sequenced last and scoped to degrade gracefully (REQ-73). Feasible **not at all** if C2 is attempted first.
Economic	**High.**	All dependencies are permissively licensed and free (CON-3); datasets are publicly available; compute is available.
Operational	**High.**	The tool slots into an existing workflow gap without requiring users to change anything else.

Conclusion. The project is feasible, **but its risk is not uniformly distributed.** It is concentrated almost entirely in one component (C2), and the schedule survives only if the process model isolates that risk rather than allowing it to block delivery of everything else. This is the decisive input to §1.7.

1.6.8 Consolidated Problem Statement

> Modern AI systems depend on datasets that frequently suffer class imbalance, demographic bias, duplicate samples, poor image quality, inconsistent labels, and insufficient diversity — defects that typically remain undetected until after training, producing inaccurate, unfair, and unreliable models.
>
> Existing tools address these dimensions **in isolation**, on **incommensurable scales**, and **without predicting consequence**. As a result, defects arising from the *interaction* of two dimensions are structurally invisible, no principled remediation priority can be established, and the practitioner has no basis on which to judge whether an audit is worth its cost.
>
> This project develops the **Dataset Integrity Analyzer**: a unified framework that evaluates all six dimensions of dataset health over a **single shared representation**, detects **cross-dimensional confounds** that no single-axis tool can observe, condenses the findings into **one interpretable AI Readiness Score**, **predicts the downstream degradation** the measured defects will induce, and issues **prioritised, actionable, interaction-aware recommendations** — all *before* model development begins.

1.7 Software Process Model Selection

The choice of process model is a requirement-bearing decision: it determines the order in which the features of Section 4 are built, when they are validated, and what happens when a component proves harder than anticipated. This section evaluates the candidates against the characteristics of *this* project and justifies the selection.

1.7.1 Project Characteristics Driving the Decision

The selection is driven by six properties established in §1.6:

#	Characteristic	Source
CH-1	**Requirements are stable at the core, uncertain at the frontier.** P1–P5 are well-defined and settled. C2's precise form (which perturbations, which surrogate, which regressor) *cannot* be specified in advance — it depends on experimental outcomes not yet observed.	§1.6.3, Appendix C (7 open TBDs)

#	Characteristic	Source
CH-2	**Risk is concentrated, not distributed.** ~80 % of technical risk sits in one component (C2); the rest is low-risk library composition.	§1.6.7
CH-3	**The architecture is naturally modular and layered.** Six analyzers over a shared substrate, then scoring, then reporting. Each dimension is independently buildable and independently demonstrable.	§2.1, NFR-Q8
CH-4	**The developer is a novice in the highest-risk area.** Learning is a project activity, not merely a precondition. Estimates for C2 are therefore unreliable.	§1.6.6
CH-5	**Partial systems have real value.** A tool that assesses five of six dimensions is genuinely useful and fully demonstrable. There is no all-or-nothing threshold.	§1.6.3
CH-6	**A hard, immovable deadline exists**, with a mandatory demonstrable artifact.	§1.6.6

1.7.2 Evaluation of Candidate Process Models

Waterfall.
Sequential, with each phase completed and frozen before the next begins.
Assessment: **Rejected.** Waterfall's precondition is that requirements can be fully and correctly specified before construction. **CH-1 violates this precondition outright** — the seven open TBDs in Appendix C are not oversights to be closed by more analysis; several (TBD-2, TBD-3, TBD-5, TBD-7) are *empirical* questions answerable only by building and measuring. Worse, Waterfall defers all integration and validation to the end, which under **CH-2** means the project's dominant risk is discovered at the point of least remaining schedule — the classic failure mode. Waterfall also delivers nothing until it delivers everything, contradicting **CH-5** and **CH-6**.

V-Model.
Waterfall with verification and validation planned in parallel with each phase.
Assessment: **Rejected**, though its testing discipline is *adopted selectively*. The V-Model inherits Waterfall's fatal assumption of up-front requirement completeness and is therefore defeated by **CH-1** for the same reason. Its emphasis on defining acceptance criteria alongside each requirement is nonetheless valuable and is retained — this is why NFR-Q7 demands that every one of REQ-1 … REQ-96 trace to an automated test case, and why NFR-Q1 mandates synthetic fixtures with analytically known ground truth. *We adopt the V-Model's rigour without its rigidity.*

Big Bang / Code-and-Fix.
Minimal process; build and see.
Assessment: **Rejected.** Under **CH-4**, the absence of structure is not liberating but hazardous:

a novice with no plan will predictably start with C2 (the interesting part), fail to debug an unfamiliar training pipeline, and arrive at **CH-6**'s deadline with nothing demonstrable. This is precisely the failure mode the chosen model is designed to prevent.

Pure Agile / Scrum.
Short timeboxed sprints, continuous re-prioritisation, working software each iteration.
Assessment: **Rejected as the governing model**, though its *practices* are adopted. Scrum presupposes a multi-person team, a Product Owner distinct from the developer, and requirements that emerge from *stakeholder negotiation*. Here there is a single developer, and the uncertainty is **technical** (will the surrogate pipeline converge?) rather than **market** (do users want this?). Scrum's ceremonies would impose overhead without addressing **CH-2**: nothing in Scrum systematically forces the highest-risk item to be confronted first. Its useful practices — incremental delivery, a working artifact at every checkpoint, backlog re-prioritisation — are inherited by the chosen model.

Incremental Model.
The system is built and delivered in successive functional increments.
Assessment: **Strong fit — adopted as the delivery mechanism. CH-3** and **CH-5** make the architecture natural to increment, and **CH-6** rewards always having something to show. But incrementality *alone* does not tell the developer **which increment to build first** — and under **CH-2** and **CH-4**, that ordering decision is the single most consequential choice in the project. Incremental is necessary but not sufficient.

Spiral Model (Boehm).
Risk-driven. Each cycle: determine objectives → **identify and analyse risks** → build and evaluate a prototype to *retire* the dominant risk → plan the next cycle.
Assessment: **Strong fit — adopted as the governing philosophy.** Spiral's organising principle is that **the highest risk is confronted earliest**, through a prototype built specifically to resolve it. This directly addresses **CH-2** and **CH-4**: C2's feasibility becomes a question answered by a cheap, early, throwaway experiment rather than a bet resolved at the deadline. Spiral also *expects* requirements to be refined between cycles — which is exactly what Appendix C's TBDs demand (**CH-1**). Its acknowledged weakness is process overhead, which is disproportionate for a solo, single-semester project if applied in full ceremony.

1.7.3 Selected Process Model

> **The project shall follow an INCREMENTAL model governed by a SPIRAL, risk-driven prioritisation of increments — that is, a Risk-Driven Incremental process.**
>
> Increments are ordered by **descending risk-retirement value**, not by architectural convenience or by developer interest. Each increment terminates in a working, demonstrable, tested artifact.

Neither model suffices alone, and the deficiency of each is the strength of the other:

- **Incremental supplies the delivery discipline** — always a working system, value early, graceful degradation if the schedule contracts (**CH-3**, **CH-5**, **CH-6**).
- **Spiral supplies the ordering discipline** — build the frightening thing early enough to survive its failure, and let each cycle refine the requirements the next cycle depends on (**CH-1**, **CH-2**, **CH-4**).

The synthesis inherits the V-Model's test-alongside-requirements rigour (NFR-Q7) and Agile's short feedback loops, while rejecting Waterfall's freeze-then-build assumption, which **CH-1** makes untenable.

1.7.4 Justification Matrix

Candidate models scored against the six driving characteristics (✓ = handles well, ~ = partially, ✗ = fails):

Characteristic	**Waterfall**	**V-Model**	**Big Bang**	**Agile/Scrum**	**Incremental**	**Spiral**	****Selected (Risk-Driven Incremental)****
CH-1 Requirements uncertain at frontier	✗	✗	~	✓	~	✓	✓
CH-2 Risk concentrated in one component	✗	✗	✗	~	~	✓	✓
CH-3 Modular, layered architecture	~	~	✗	✓	✓	✓	✓
CH-4 Novice developer; unreliable estimates	✗	✗	✗	~	✓	✓	✓
CH-5 Partial system has real value	✗	✗	✗	✓	✓	✓	✓
CH-6 Hard deadline; artifact mandatory	~	~	✗	✓	✓	~	✓

Characteristic	Waterfall	V-Model	Big Bang	Agile/Scrum	Incremental	Spiral	**Selected (Risk-Driven Incremental)**
Overhead appropriate for solo developer	✓	~	✓	~	✓	✗	✓

The selected model is the only column without a ✗ — it takes Spiral's risk ordering while shedding the ceremony that makes full Spiral inappropriate at this scale (final row).

1.7.5 Increment Plan

Increments are ordered by risk retirement, **not** by the section order of this document. Note in particular that **Increment 1 is a throwaway spike**, and that **C2 — the project's most novel component — is scheduled fourth, not first.**

#	Increment	Risk retired	Deliverable	Exit criterion
0	**Skeleton + ingestion + one dimension** (§4.1, §4.3)	*Can I build anything end-to-end?*	Folder in → class-distribution report out	Runs on UTKFace; imbalance statistics correct against a hand-computed fixture
1	**C2 FEASIBILITY SPIKE** *(throwaway)*	**The dominant risk (CH-2).** *Can I inject a known bias, train a surrogate, and observe the predicted degradation — at all?*	A single script: perturb CIFAR-10 at 3 imbalance levels, train ResNet-18 on each, plot accuracy vs. imbalance	**A monotone curve is observed.** If not → C2 is descoped to correlational analysis and the project proceeds on C1 alone, **with ~85 % of the schedule still intact**
2	**Remaining dimensions** (§4.2, §4.4–§4.7)	*Integration risk* — do six analyzers share one substrate cleanly?	All six dimensions writing to the unified per-sample record (REQ-53)	Every dimension unit-tested against synthetic fixtures (NFR-Q1)
3	**C1 — confound analysis + scoring + report** (§4.8, §4.10, §4.12)	*Novelty risk* — **is the primary contribution real?**	Cross-dimensional confounds detected; AI Readiness Score; exportable report	A **deliberately injected** confound (quality degraded only for one sub-group) is successfully detected

#	Increment	Risk retired	Deliverable	Exit criterion
4	**C2 — full bias impact prediction** (§4.9)	*(Largely retired in Increment 1)*	Trained predictor + LODO validation + ablation (REQ-69)	Predictor beats a mean baseline on held-out datasets
5	**Recommendations, UI, hardening** (§4.11, §3.1, §5)	*Usability risk*	Dashboard, ranked recommendations, Docker image	Documented NFRs met

The critical design property of this plan is Increment 1. It is deliberately *cheap* (one dataset, one architecture, three data points), deliberately *early*, and deliberately *disposable*. Its sole function is to convert the project's dominant unknown into a known while the schedule can still absorb the answer. Under Waterfall that question would be answered in the final weeks; under Big Bang it would consume the entire semester; here it is answered in days.

This is the concrete operational meaning of "risk-driven," and it is the single most important reason for the selection.

1.7.6 Requirements Traceability to the Process Model

The selected model is not merely a schedule — it is *encoded as requirements* elsewhere in this document, and those requirements exist **because** of the process choice:

Requirement	Encodes
REQ-73 — the system shall degrade gracefully if the Bias Impact Predictor is unavailable; all other features remain fully functional	**The Increment-1 abort path.** If the spike fails, the product is still a coherent, deliverable system. This requirement is what *makes* C2 safely descopable, and it exists precisely because the process model demands the option.
NFR-Q8 — each dimension is an independently replaceable module writing to the shared record	**Incrementality made architectural.** Increment 2 adds dimensions without modifying the orchestrator, scorer, or confound engine.
NFR-Q7 — every REQ traces to an automated test	**The V-Model discipline retained.** Each increment is verifiable at its exit criterion rather than at project end.
NFR-Q1 — metrics validated against analytically known synthetic fixtures	**Exit criteria made objective.** "Increment 2 is done" is a test result, not a judgement call.

Requirement	Encodes
Appendix C (TBD-2, -3, -4, -5, -7) — open items with named resolution points	**Spiral's between-cycle refinement.** These are not defects in the specification; they are questions the process model *schedules* rather than guesses at.

That the SRS already contains REQ-73 and NFR-Q8 is not coincidental. They were written *because* the process model requires the system to survive the failure of its most ambitious component — which is the clearest available evidence that the process model was selected before, and shaped, the requirements it governs.

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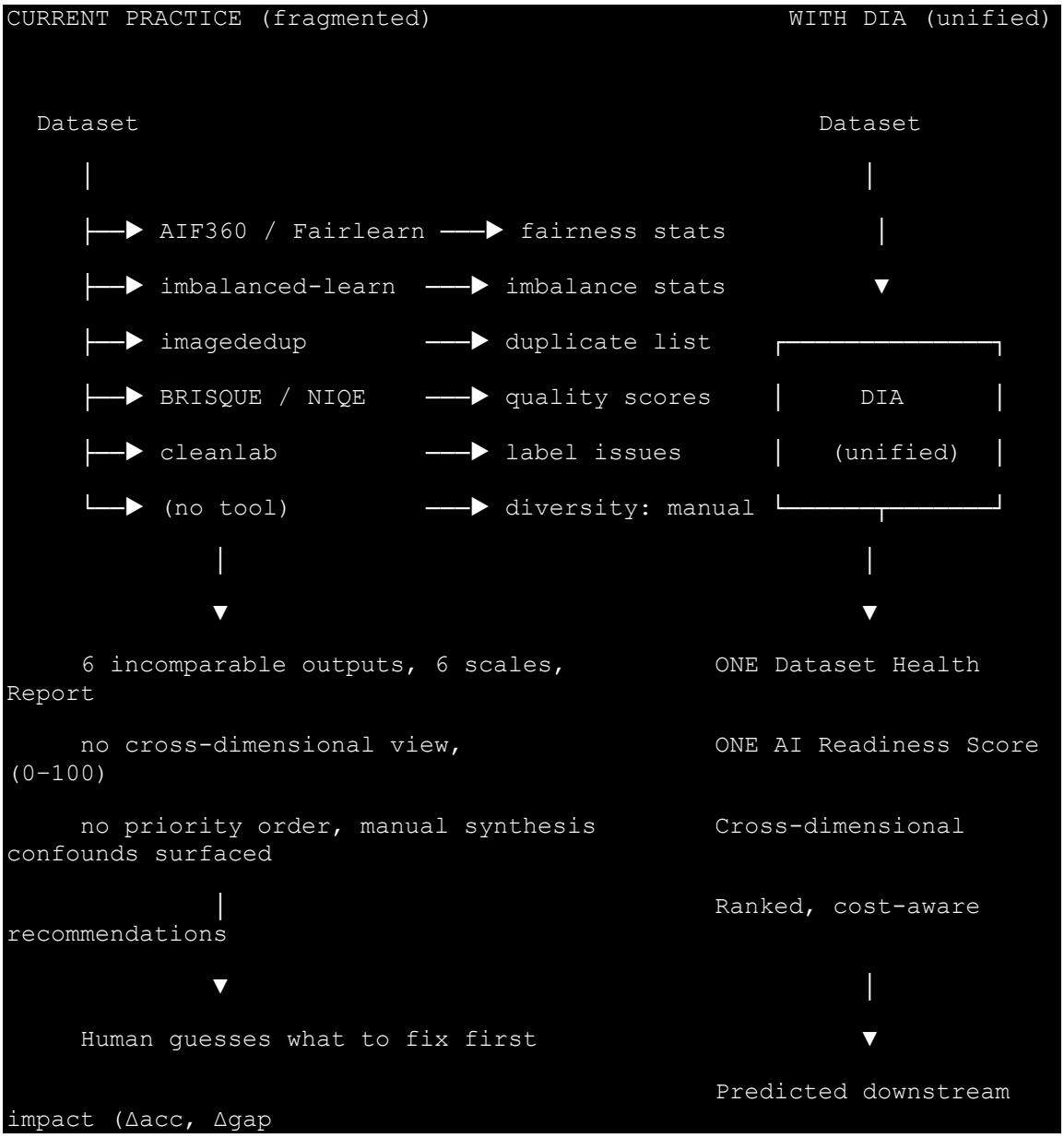
2. Overall Description

2.1 Product Perspective

Dataset Integrity Analyzer is a **new, self-contained product**. It is not an extension of any single existing system; it is a **consolidation** of a class of them.

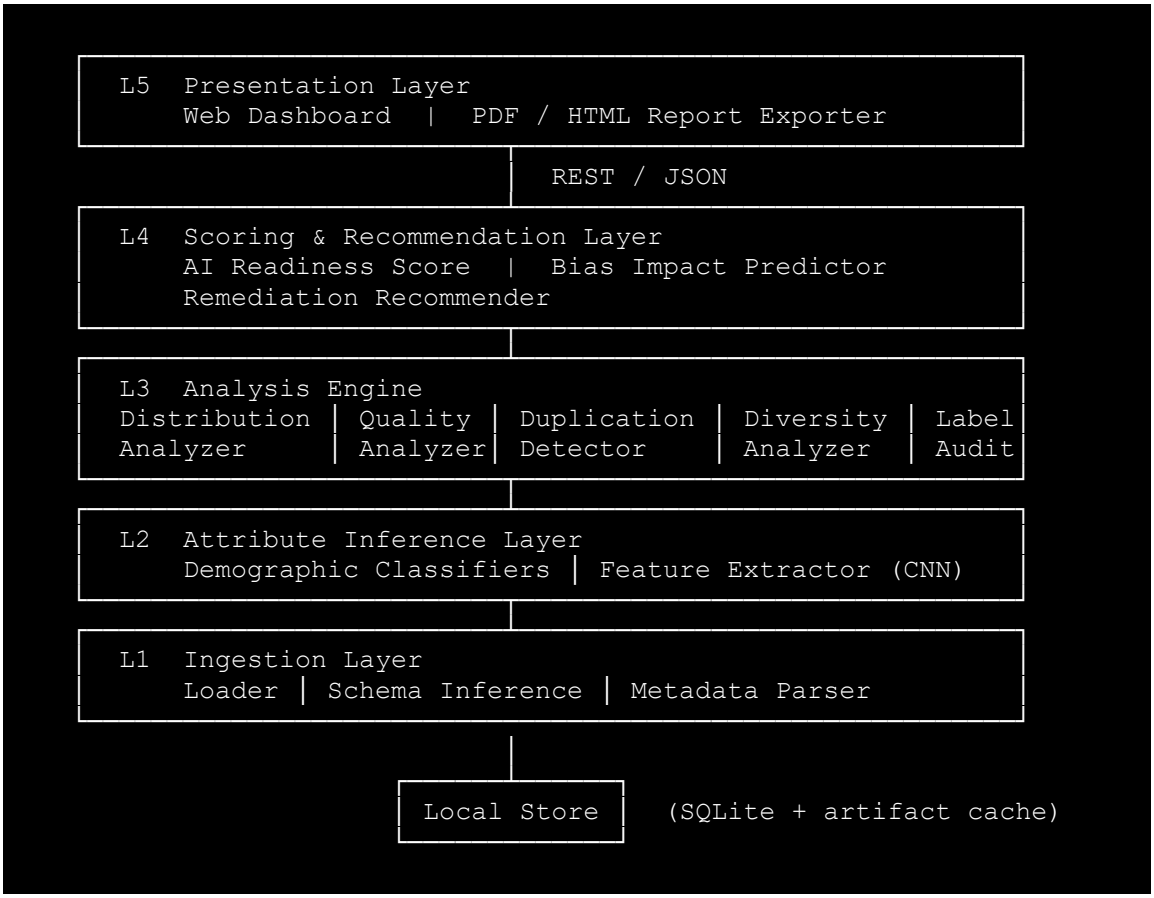
DIA occupies a position *upstream* of the conventional ML pipeline: where standard practice moves directly from *data collection* to *model training*, DIA inserts an explicit **dataset validation gate** between them.

Relation to existing systems. DIA subsumes the *diagnostic* function of the fragmented single-axis toolchain that a rigorous practitioner must otherwise assemble by hand:



Critically, DIA is **not** a wrapper that shells out to those tools and concatenates their outputs. Such a wrapper would inherit the fragmentation it purports to solve: six independent passes over the data, six sample indices that cannot be joined, and no shared representation over which a cross-dimensional confound could be computed. Instead, DIA performs **one ingestion pass and one feature-extraction pass**, and all six analysis dimensions operate over that **single shared sample index and single shared embedding store** (D1 and D2 in Appendix B.2). This shared substrate is the architectural precondition for every capability claimed in §1.4.3: it is what allows the Quality analyzer's per-sample scores to be regressed against the Attribute analyzer's per-sample demographic labels (the quality-confound test, REQ-54), and it is what allows the bias descriptor vector (REQ-64) to be assembled at all.

The system is architected as five loosely coupled layers:



External interfaces are limited to: (a) the user's local or mounted dataset directory, (b) the browser, and (c) optionally, pre-trained model weights fetched once from a model hub.

2.2 Product Functions

At a high level, the product shall:

- **F1 — Ingest** an image dataset supplied as a folder hierarchy, ZIP archive, or manifest file (CSV/JSON), and infer its schema (classes, splits, available metadata).
- **F2 — Infer attributes** (gender, age band, skin tone / lighting condition) for samples where such metadata is absent, using pre-trained classifiers.
- **F3 — Analyse class and demographic distribution**, computing imbalance ratios, entropy, and disparity statistics.

- **F4 — Assess image quality** per sample (blur, brightness, contrast, noise, resolution, corruption) and aggregate to dataset level.
- **F5 — Detect duplicates and near-duplicates** via perceptual hashing and embedding similarity, including cross-split leakage.
- **F6 — Measure diversity and coverage** through embedding-space clustering, identifying redundant clusters and under-represented regions.
- **F7 — Audit label consistency**, flagging probable mislabels and outliers.
- **F8 — Detect cross-dimensional confounds** by joining all of the above on a single shared sample index — quality × demographic, quality × class, redundancy × group, diversity × class, label-noise × group, leakage × class. *These defects are structurally invisible to any single-axis tool.*
- **F9 — Predict bias impact**: estimate the downstream accuracy drop and sub-group fairness gap that the detected defects, and their interactions, will induce.
- **F10 — Compute an overall dataset integrity score (the AI Readiness Score)** as an interpretable weighted composite that places all six dimensions on **one common, comparable 0–100 scale**.
- **F11 — Recommend remediations**, prioritised across dimensions by predicted benefit per unit effort, and made interaction-aware where a confound is present.
- **F12 — Generate a unified Dataset Health Report** — one report covering all six dimensions, viewable in-browser and exportable to PDF/HTML/JSON.

Functions **F8**, **F10**, and **F11** are each *only possible under unification*; a composition of independent single-axis tools cannot provide any of them (see §1.4.1).

2.3 User Classes and Characteristics

User Class	Characteristics	Frequency	Importance
ML Researcher / Data Scientist *(primary)*	Strong Python and ML background; needs quantitative, defensible metrics and exportable artifacts for papers. Uses all functions.	High	**Critical**
ML Practitioner / Engineer *(primary)*	Applied focus; wants a pass/fail readiness verdict and a concrete fix list before committing GPU budget. Uses F1, F9, F10, F11 most.	High	**Critical**
Student / Learner *(secondary)*	Limited ML depth; relies on plain-language explanations and visualisations. Needs guided defaults.	Medium	Important

User Class	Characteristics	Frequency	Importance
Auditor / Compliance Reviewer *(secondary)*	Non-technical or semi-technical; consumes only the exported Dataset Health Report (F11) and fairness findings (F3, F8). Does not run analyses.	Low	Important
System Administrator *(tertiary)*	Deploys and maintains the application; configures thresholds, model weights, and resource limits.	Very Low	Minor

The **ML Researcher** and **ML Practitioner** classes are the most important to satisfy; where their needs conflict with other classes, they take precedence.

2.4 Operating Environment

- **OE-1:** The system shall operate on a 64-bit Linux (Ubuntu 20.04 LTS or later), Windows 10/11, or macOS 12+ host.
- **OE-2:** The system shall execute on **Python 3.10 or later**.
- **OE-3:** The system shall function on CPU-only hardware with a minimum of 8 GB RAM; it shall exploit a CUDA-capable GPU (compute capability ≥ 6.0) when available to accelerate feature extraction.
- **OE-4:** The web dashboard shall render correctly on current versions of Chrome, Firefox, and Edge at a minimum viewport width of 1280 px.
- **OE-5:** The system shall require a minimum of 5 GB free disk space for the artifact cache and embedding store.
- **OE-6:** The system shall be deployable as a Docker container to guarantee environment reproducibility.
- **OE-7:** The system shall operate fully offline after first-run model weight download.

2.5 Design and Implementation Constraints

- **CON-1:** Implementation language shall be Python 3.10+; the back-end shall use FastAPI (or Flask) and the front-end React (or Streamlit for the reduced-effort variant).
- **CON-2:** Deep learning components shall use PyTorch ≥ 2.0 with torchvision.
- **CON-3:** All third-party libraries shall carry permissive licences (MIT, BSD, Apache-2.0) to permit publication of the source alongside the conference paper.
- **CON-4:** The user's dataset shall never be modified, moved, or deleted — the system opens all dataset files read-only.
- **CON-5:** No dataset content, image, or derived embedding shall be transmitted off-host. All computation is local.

- **CON-6:** Persistence shall use SQLite (single-file, zero-configuration) rather than a server-based RDBMS.
- **CON-7:** Demographic attribute inference is inherently approximate; every inferred attribute shall be stored and displayed with an associated confidence value and shall never be presented as ground truth.
- **CON-8:** The Bias Impact Prediction model shall be trainable within the project's compute budget (single consumer GPU, ≤ 48 h total), constraining the surrogate-model architecture to lightweight CNNs (e.g. ResNet-18, MobileNetV3).
- **CON-9:** Peak memory usage shall be bounded by streaming/batched processing; the system shall never require the full dataset to be resident in RAM.

2.6 User Documentation

The following shall be delivered alongside the software:

- **DOC-1:** A `README.md` covering installation, Docker usage, and a five-minute quick-start.
- **DOC-2:** A User Guide (PDF) describing dataset layout requirements, the analysis workflow, and interpretation of each metric in the Dataset Health Report.
- **DOC-3:** API reference documentation, auto-generated from docstrings (Sphinx) and, for the REST layer, an OpenAPI/Swagger specification served at `/docs`.
- **DOC-4:** An annotated worked example (Jupyter notebook) running the full pipeline on the UTKFace dataset.
- **DOC-5:** A metric reference sheet defining each formula, its range, and its interpretation thresholds.

2.7 Assumptions and Dependencies

Assumptions

- **AS-1:** The dataset supplied is an *image classification* dataset with a discoverable class structure (folder-per-class or manifest column).
- **AS-2:** Images are in a standard raster format (JPEG, PNG, BMP, TIFF) readable by Pillow/OpenCV.
- **AS-3:** The user has read access to the dataset directory and has the right to analyse it.
- **AS-4:** Pre-trained demographic classifiers trained on FairFace generalise acceptably to the user's face-image datasets. This assumption does not hold for non-face domains (e.g. chest X-rays), where demographic analysis is disabled and only quality/duplication/diversity dimensions apply.

- **AS-5:** For the Bias Impact Prediction module, the relationship learned between dataset-level bias descriptors and downstream performance degradation is assumed to transfer across datasets of similar modality.

Dependencies

- **DE-1:** Availability of PyTorch, torchvision, scikit-learn, OpenCV, Pillow, imagehash, NumPy, pandas, SciPy, UMAP/t-SNE, and a plotting library (Plotly/Matplotlib).
- **DE-2:** One-time network access to fetch pre-trained model weights (ImageNet-pretrained backbone; FairFace-trained demographic heads).
- **DE-3:** Availability of public benchmark datasets (UTKFace, FairFace, CelebA, HAM10000, Chest X-Ray, CIFAR-10/100, COCO/ImageNet subsets) from Kaggle, GitHub, or IEEE DataPort for training and validating the Bias Impact Predictor.
- **DE-4:** A modern browser for the dashboard.

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3. External Interface Requirements

3.1 User Interfaces

- **UI-1:** The system shall present a browser-based dashboard with five primary screens: **Upload/Configure**, **Analysis Progress**, **Health Dashboard**, **Detailed Findings**, and **Recommendations**.
- **UI-2:** The **Upload/Configure** screen shall accept a directory path, ZIP upload, or manifest file, and shall allow the user to enable/disable individual analysis dimensions and to set the sensitive-attribute set to analyse.
- **UI-3:** The **Analysis Progress** screen shall display a per-stage progress bar, elapsed time, estimated remaining time, and a cancel control.
- **UI-4:** The **Health Dashboard** shall display the AI Readiness Score prominently as a 0–100 gauge with a colour band (Red $< 40 \leq$ Amber $< 70 \leq$ Green), together with a radar chart of the six dimension sub-scores and a ranked list of critical issues.
- **UI-5:** The **Detailed Findings** screen shall provide, per dimension, the computed statistics, supporting charts (histograms, heatmaps, 2-D embedding scatter), and a thumbnail gallery of exemplar offending samples (e.g. detected duplicates, blurred images, suspected mislabels).
- **UI-6:** The **Recommendations** screen shall list each recommended action with its target defect, predicted score improvement, and estimated effort.
- **UI-7:** Every screen shall provide a persistent **Export Report** control offering PDF, HTML, and JSON formats.

- **UI-8:** All charts shall be accompanied by textual alt-text descriptions, and colour shall never be the sole carrier of meaning (WCAG 2.1 AA colour-independence).
- **UI-9:** Error conditions (unreadable file, empty class, unsupported format) shall be surfaced inline with a plain-language message and a suggested corrective action, and shall not terminate the analysis of unaffected dimensions.
- **UI-10:** A command-line interface (`dia analyze --path <dir> --out <report>`) shall expose the full pipeline for headless/batch use.

3.2 Hardware Interfaces

- **HI-1:** The system shall interface with the host file system for read-only dataset access and read-write artifact cache access via standard OS calls.
- **HI-2:** The system shall interface with a CUDA-capable GPU, where present, through the PyTorch CUDA runtime, and shall transparently fall back to CPU tensors when no compatible device is detected.
- **HI-3:** No specialised, custom, or proprietary hardware is required. No direct device-driver-level interaction is performed.

3.3 Software Interfaces

Component	Version	Purpose	Data In / Out
Python runtime	≥ 3.10	Execution environment	—
PyTorch / torchvision	≥ 2.0	CNN feature extraction, demographic inference, surrogate-model training	In: image tensors. Out: embeddings (`float32[N,D]`), class logits
OpenCV / Pillow	≥ 4.8 / ≥ 10.0	Image decoding, blur/brightness/contrast/noise metrics	In: file bytes. Out: `ndarray`, scalar quality metrics
imagehash	≥ 4.3	Perceptual hashing (pHash/dHash) for exact and near-duplicate detection	In: PIL image. Out: 64-bit hash
scikit-learn	≥ 1.3	Clustering (k-means, DBSCAN), PCA, isolation forest, surrogate regressor	In: embeddings. Out: cluster labels, anomaly scores, predictions

Component	Version	Purpose	Data In / Out
SciPy / NumPy / pandas	current	Statistical tests, entropy, imbalance ratios, tabular aggregation	In/Out: arrays, dataframes
UMAP	≥ 0.5	2-D projection of embedding space for diversity visualisation	In: `float32[N,D]`. Out: `float32[N,2]`
FastAPI + Uvicorn	≥ 0.110	REST API and static asset serving	In: HTTP/JSON. Out: HTTP/JSON
SQLite	≥ 3.35	Persistence of analysis runs, per-sample metrics, and reports	In/Out: SQL
ReportLab / WeasyPrint	current	PDF report rendering	In: HTML/report model. Out: PDF bytes
Plotly / Matplotlib	current	Chart generation for dashboard and report	In: metric arrays. Out: SVG/PNG/JSON figures

Shared data across components is the **Analysis Run** object (run identifier, dataset fingerprint, per-sample metric table, per-dimension aggregate, score, recommendation list), persisted in SQLite and serialisable to JSON. This object is the single source of truth for all downstream consumers (dashboard, exporter, CLI).

3.4 Communications Interfaces

- **CI-1:** The dashboard shall communicate with the back-end exclusively over **HTTP/1.1** using **JSON** payloads (`application/json`), conforming to a documented OpenAPI 3.0 specification.
- **CI-2:** Long-running analysis progress shall be streamed to the client via **Server-Sent Events (SSE)** on `GET /runs/{run_id}/events`, avoiding polling.
- **CI-3:** File upload shall use `multipart/form-data` with a configurable maximum body size (default 2 GB).
- **CI-4:** When bound to a non-loopback interface, the server **shall** be served over **HTTPS/TLS 1.2+**; the default configuration binds to `127.0.0.1` only.
- **CI-5:** The only outbound communication permitted is a one-time HTTPS fetch of pre-trained model weights from a configured model-hub URL. No telemetry, analytics, or dataset content is ever transmitted.
- **CI-6:** No email, message-queue, or third-party API integration is required in Version 1.0.

4. System Features

Functional requirements are organised by system feature — the major services the product provides. Priorities are stated per feature and inherited by contained requirements unless overridden.

4.1 Dataset Ingestion and Schema Inference

4.1.1 Description and Priority

Accepts a dataset in a supported layout, validates it, and infers its schema (classes, splits, metadata columns, sample count). All other features depend on this one.
Priority: High *(Benefit 9, Penalty 9, Cost 3, Risk 3)*

4.1.2 Stimulus/Response Sequences

1. User selects a dataset path / uploads a ZIP / supplies a manifest → System validates readability and reports the discovered structure.

2. System encounters an unreadable or corrupt image → System logs it, excludes it from analysis, and increments a `corrupt_files` counter shown in the report.

3. System detects no discoverable class structure → System prompts the user to supply a manifest mapping filenames to labels.

4.1.3 Functional Requirements

- **REQ-1:** The system shall accept datasets in (a) folder-per-class layout, (b) `split/class/` nested layout, and (c) flat directory with an accompanying CSV or JSON manifest.
- **REQ-2:** The system shall accept a ZIP archive and extract it to a temporary working directory, deleting the extraction on run completion.
- **REQ-3:** The system shall enumerate all files with extensions `.jpg`, `.jpeg`, `.png`, `.bmp`, `.tif`, `.tiff` (case-insensitive) and treat all others as non-image metadata.
- **REQ-4:** The system shall infer and record: total sample count, class list, per-class count, split membership (if present), and any additional metadata columns present in a manifest.
- **REQ-5:** The system shall parse standard metadata attribute columns where present (e.g. `age`, `gender`, `race`, `ethnicity`), including UTKFace-style filename-encoded attributes (`<age>_<gender>_<race>_<timestamp>.jpg`).
- **REQ-6:** The system shall compute a stable dataset fingerprint (hash over the sorted list of file paths and sizes) to permit caching and re-run detection.
- **REQ-7:** If a file cannot be decoded, the system shall record it as corrupt, exclude it from all downstream analysis, and continue without aborting the run.

- **REQ-8:** If fewer than 2 classes are discovered, the system shall reject the dataset with the message "Dataset must contain at least two classes for integrity analysis."
- **REQ-9:** The system shall support a user-specified sampling rate (default: analyse all samples; configurable to analyse a random *n*-sample subset for very large datasets), and shall record the sampling rate in the report.

4.2 Attribute Inference Engine

4.2.1 Description and Priority

Infers sensitive/protected attributes (gender, age band, skin-tone band) and image-condition attributes (lighting) for samples lacking ground-truth metadata, so that fairness analysis is possible on unlabelled datasets.
Priority: High *(Benefit 8, Penalty 8, Cost 6, Risk 7)*

4.2.2 Stimulus/Response Sequences

1. Ingestion completes with no demographic metadata → System runs the pre-trained attribute classifiers over all samples and populates inferred attribute columns with confidence values.

2. Ground-truth attributes are present in the manifest → System uses ground truth and skips inference, flagging in the report that attributes are authoritative.

3. Dataset is detected as non-face (no face detected in a sampled subset) → System disables demographic inference and records that fairness-by-demographic analysis is not applicable.

4.2.3 Functional Requirements

- **REQ-10:** The system shall infer **gender** (binary predicted category with confidence), **age band** (0–12, 13–19, 20–29, 30–39, 40–49, 50–59, 60+), and **skin-tone band** (Fitzpatrick-approximate 6-level or a coarse 3-level proxy — see **TBD-1**) for each face-containing sample.
- **REQ-11:** The system shall compute a **lighting condition** label (Dark / Normal / Bright) from the mean luminance of the image in the Lab colour space, using configurable thresholds.
- **REQ-12:** Each inferred attribute shall be stored with a confidence score in `[0,1]` and shall be excluded from aggregate statistics when confidence falls below a configurable threshold (default 0.60); the count of low-confidence exclusions shall be reported.
- **REQ-13:** When ground-truth attributes are supplied, the system shall prefer them over inferred values and shall state this provenance explicitly in the Dataset Health Report.
- **REQ-14:** The system shall run a face detector on a random sample of ≥ 200 images (or the whole dataset if smaller); if the face-detection rate is below 20 %, the system shall classify the dataset as non-face and disable REQ-10.

- **REQ-15:** The Dataset Health Report shall carry a mandatory, non-suppressible disclaimer stating that inferred demographic attributes are model estimates, are themselves subject to error and bias, and must not be treated as ground truth about individuals.
- **REQ-16:** The system shall report the attribute classifiers' own validation accuracy (per class, on a held-out FairFace split) so the user can calibrate trust in the inferred distributions.

4.3 Class and Demographic Distribution Analysis

4.3.1 Description and Priority

Quantifies imbalance across target classes and across sensitive attribute groups, including the intersection of the two.
Priority: High *(Benefit 9, Penalty 9, Cost 3, Risk 2)*

4.3.2 Stimulus/Response Sequences

1. Attributes available → System computes per-class counts, per-attribute counts, and the class × attribute contingency table.

2. Imbalance Ratio exceeds a severity threshold → System raises a **Critical Issue** entry visible on the Health Dashboard.

3. A class × attribute cell has zero or near-zero support → System raises a **Coverage Gap** issue naming the missing sub-group.

4.3.3 Functional Requirements

- **REQ-17:** The system shall compute, for the target class variable: absolute counts, relative frequencies, **Imbalance Ratio** IR = n(majority)/n(minority), normalised **Shannon entropy** H/log k, and the **Gini coefficient**.
- **REQ-18:** The system shall compute the same statistics for each sensitive attribute (gender, age band, skin tone, lighting).
- **REQ-19:** The system shall construct the **class × sensitive-attribute contingency table** and compute a **χ^2 test of independence**, reporting the statistic, degrees of freedom, and p-value, together with **Cramér's V** as an effect size.
- **REQ-20:** The system shall compute the **Demographic Parity Difference** and **Disparate Impact Ratio** of representation across each sensitive group.
- **REQ-21:** The system shall identify and enumerate **intersectional sub-groups** (e.g. *female × age 60+ × dark lighting*) whose sample count falls below a configurable minimum (default: 1 % of dataset or 50 samples, whichever is greater) and flag them as coverage gaps.
- **REQ-22:** The system shall classify imbalance severity as **Balanced** (IR ≤ 1.5), **Mild** (1.5 < IR ≤ 3), **Moderate** (3 < IR ≤ 10), or **Severe** (IR > 10), and shall surface Severe findings as Critical Issues.

- **REQ-23:** The system shall produce, for the report, a per-class bar chart, per-attribute pie/bar charts, and a class × attribute heatmap.
- **REQ-24:** Where a split (train/val/test) is present, the system shall additionally compute the distribution divergence between splits using **Jensen–Shannon divergence** and flag splits whose divergence exceeds a configurable threshold (default 0.1) as a **Split Skew** issue.

4.4 Image Quality Assessment

4.4.1 Description and Priority

Computes per-sample no-reference image quality metrics and aggregates them to identify systematic quality defects and quality–subgroup correlations.
Priority: High *(Benefit 8, Penalty 7, Cost 4, Risk 3)*

4.4.2 Stimulus/Response Sequences

1. Analysis starts → System computes blur, brightness, contrast, noise, resolution, and aspect-ratio metrics per sample.

2. A sample falls below a quality threshold on any metric → System tags it and includes it in the low-quality exemplar gallery.

3. Low-quality samples concentrate in one class or one demographic sub-group → System raises a **Quality Confound** issue, because such a correlation is a covert source of model bias.

4.4.3 Functional Requirements

- **REQ-25:** The system shall compute per sample: **blur score** (variance of the Laplacian), **brightness** (mean Lab L*), **contrast** (standard deviation of L*), **noise estimate** (median absolute deviation of a high-pass residual), **resolution** (W × H), **aspect ratio**, and **file size**.
- **REQ-26:** The system shall compute a no-reference perceptual quality score per sample (BRISQUE or equivalent) and normalise it to `[0,100]`.
- **REQ-27:** The system shall flag samples as **Blurred** (Laplacian variance below a configurable threshold, default 100), **Underexposed** (mean $L^* < 40$), **Overexposed** (mean $L^* > 215$), **Low-contrast** ($\sigma(L^*) < 20$), or **Under-resolution** (min side < 32 px).
- **REQ-28:** The system shall aggregate flagged proportions per quality defect and report them as percentages of the analysed dataset.
- **REQ-29:** The system shall test, for each quality metric, whether its distribution differs significantly across target classes and across sensitive groups (Kruskal–Wallis H test, $\alpha = 0.05$), and shall raise a **Quality Confound** issue where a significant difference with non-trivial effect size is found.
- **REQ-30:** The system shall detect **corrupt or truncated images** (decode failure, all-black, all-white, or zero-variance frames) and enumerate them separately from quality-flagged images.

- **REQ-31:** The system shall present a histogram per quality metric and a per-class box plot in the Detailed Findings screen.

4.5 Duplicate and Near-Duplicate Detection

4.5.1 Description and Priority

Identifies exact duplicates, perceptual near-duplicates, and — critically — duplicates that straddle the train/test boundary, which silently inflate reported model accuracy.
Priority: High *(Benefit 8, Penalty 9, Cost 4, Risk 3)*

4.5.2 Stimulus/Response Sequences

1. Analysis starts → System computes a cryptographic hash and a perceptual hash for every sample.

2. Two samples share an identical cryptographic hash → System records an **Exact Duplicate** pair.

3. Two samples have perceptual-hash Hamming distance ≤ threshold, or embedding cosine similarity ≥ threshold → System records a **Near-Duplicate** pair.

4. A duplicate pair spans train and test splits → System raises a **Data Leakage** Critical Issue.

4.5.3 Functional Requirements

- **REQ-32:** The system shall compute an **MD5 (or SHA-256) content hash** per image file and group identical hashes into exact-duplicate sets.
- **REQ-33:** The system shall compute a **perceptual hash** (pHash, 64-bit) per image and identify near-duplicate pairs whose Hamming distance is ≤ a configurable threshold (default 5).
- **REQ-34:** The system shall compute **CNN embedding cosine similarity** for candidate pairs and confirm near-duplicates where similarity ≥ a configurable threshold (default 0.98), reducing false positives from perceptual hashing alone.
- **REQ-35:** The system shall use an approximate-nearest-neighbour index (or LSH over the hash space) so that duplicate detection does not require an $O(N^2)$ exhaustive pairwise comparison.
- **REQ-36:** The system shall report: total exact duplicates, total near-duplicates, the number of distinct duplicate groups, and the **redundancy ratio** = duplicated samples / total samples.
- **REQ-37:** Where splits are defined, the system shall specifically enumerate **cross-split duplicate pairs** and shall raise a Critical Issue titled "Train–Test Leakage" whenever any such pair exists, quantifying the number of leaked test samples.
- **REQ-38:** The system shall detect **conflicting-label duplicates** — visually identical samples carrying different class labels — and report them as high-priority label defects.

- **REQ-39:** The Detailed Findings screen shall display side-by-side thumbnail pairs for the top *n* (default 20) duplicate groups.

4.6 Diversity and Coverage Analysis

4.6.1 Description and Priority

Measures whether the dataset spans the visual variation the model will encounter, or is instead concentrated in a few redundant modes.
Priority: Medium *(Benefit 7, Penalty 6, Cost 5, Risk 4)*

4.6.2 Stimulus/Response Sequences

1. Feature extraction completes → System clusters the embedding space and reports cluster occupancy.

2. A small number of clusters dominates → System raises a **Low Diversity** issue.

3. A class occupies very few clusters relative to others → System raises a **Class-Level Diversity Deficit** issue naming that class.

4.6.3 Functional Requirements

- **REQ-40:** The system shall extract a fixed-length embedding (e.g. 512-D from an ImageNet-pretrained ResNet-18/50 penultimate layer) for every analysed sample.
- **REQ-41:** The system shall cluster the embedding space using k-means (with *k* selected by the silhouette criterion over a configurable range) and, additionally, DBSCAN to identify density structure and outliers.
- **REQ-42:** The system shall compute dataset-level diversity as: **normalised cluster entropy**, **mean intra-class embedding variance**, and the **mean pairwise cosine distance** over a random sample of pairs.
- **REQ-43:** The system shall compute **per-class diversity** and flag any class whose diversity falls more than one standard deviation below the dataset mean.
- **REQ-44:** The system shall identify **redundant clusters** (clusters whose members exhibit intra-cluster mean cosine similarity above a configurable threshold, default 0.95) and estimate the number of samples that could be pruned without loss of coverage.
- **REQ-45:** The system shall project the embedding space to two dimensions (UMAP, with t-SNE as an alternative) and render a class-coloured scatter plot in the Detailed Findings screen.
- **REQ-46:** The system shall identify **sparse regions** of the embedding space — regions with low sample density but non-zero occupancy — and characterise them by exemplar thumbnails, as candidate under-represented conditions.

4.7 Label Consistency and Anomaly Detection

4.7.1 Description and Priority

Surfaces probable mislabels and out-of-distribution samples so the user can triage annotation errors before they propagate into training.
Priority: Medium *(Benefit 7, Penalty 7, Cost 6, Risk 5)*

4.7.2 Stimulus/Response Sequences

1. Analysis starts → System trains a lightweight cross-validated classifier on the embeddings and obtains out-of-fold predicted probabilities for every sample.

2. A sample's given label receives very low predicted probability while another class receives high probability → System flags it as a **Suspected Mislabel**.

3. A sample lies far from every class centroid → System flags it as an **Outlier / Out-of-Distribution** sample.

4.7.3 Functional Requirements

- **REQ-47:** The system shall train a *k*-fold cross-validated classifier (default: logistic regression or a small MLP over the extracted embeddings; $k = 5$) and obtain out-of-fold predicted class probabilities for every sample.
- **REQ-48:** The system shall apply a confident-learning style criterion to identify **label issues**: samples whose self-confidence (probability assigned to the given label) is below the class-specific threshold while a competing class exceeds it.
- **REQ-49:** The system shall rank suspected mislabels by a confidence margin and expose the top *n* (default 50) with thumbnails, given label, and predicted label.
- **REQ-50:** The system shall estimate the **overall label noise rate** as the proportion of samples flagged, and report it with a stated caveat that it is an estimate derived from a surrogate model, not from human re-annotation.
- **REQ-51:** The system shall detect **outliers** using Isolation Forest and/or a Mahalanobis-distance criterion in embedding space, and report the outlier proportion per class.
- **REQ-52:** The system shall report **per-class label reliability** (1 − flagged proportion for that class) so that unreliable classes can be prioritised for re-annotation.

4.8 Cross-Dimensional Confound Analysis (Novel Contribution C1)

4.8.1 Description and Priority

This feature is the concrete deliverable of the **unification contribution (C1)**. It detects defects that arise from the **interaction of two or more dimensions** and that are therefore **structurally invisible to any single-axis tool**. A fairness toolkit inspects labels and cannot see pixels; an image-quality metric inspects pixels and cannot see demographics; neither can discover that the underexposed images in a dataset are disproportionately images of dark-

skinned subjects — a defect that will produce a model that is accurate in aggregate and unfair in the particular.

This feature is only implementable because every dimension in §4.3–§4.7 writes to the **same shared sample index** (D1) and the **same shared embedding store** (D2). It is what a wrapper around six separate tools could not provide.
Priority: High *(Benefit 9, Penalty 8, Cost 5, Risk 4)*

4.8.2 Stimulus/Response Sequences

1. All single-dimension analyses (§4.3–§4.7) complete → System joins their per-sample outputs on the shared sample index and executes the full battery of cross-dimensional tests.

2. A statistically significant association is found between a quality defect and a sensitive attribute → System raises a **Quality–Fairness Confound** Critical Issue naming both the defect and the affected sub-group.

3. Duplicate samples are found to concentrate in one class or one sub-group → System raises a **Redundancy–Fairness Confound**, since the effective (deduplicated) representation of that group is lower than the raw counts suggest.

4. No significant interaction is found on any pair → System records "no cross-dimensional confounds detected" as a positive finding in the report, rather than silently omitting the section.

4.8.3 Functional Requirements

- **REQ-53:** The system shall maintain a **unified per-sample record** joining, on a single `sample_id`, every per-sample output of §4.2–§4.7: class label, inferred/ground-truth sensitive attributes with confidences, all image-quality metrics, duplicate-group membership, embedding-cluster membership, and label-issue flag. All cross-dimensional analysis shall operate on this joined record.
- **REQ-54:** The system shall test for a **Quality–Fairness Confound**: for each image-quality metric and each sensitive attribute, test whether the quality distribution differs significantly across attribute groups (Kruskal–Wallis, $\alpha = 0.05$, with Benjamini–Hochberg correction across the full family of tests), reporting the effect size (ε^2 or Cliff's δ) alongside the *p*-value.
- **REQ-55:** The system shall test for a **Quality–Class Confound**: whether image quality differs significantly across target classes, which would allow a model to exploit a quality artefact as a spurious shortcut for the label.
- **REQ-56:** The system shall test for a **Redundancy–Fairness/Class Confound**: whether duplicate and near-duplicate samples are non-uniformly distributed across classes or sensitive groups, and shall report each group's **effective sample count** (unique samples) alongside its raw count, since redundancy inflates apparent representation.
- **REQ-57:** The system shall test for a **Diversity–Class Confound**: whether per-class embedding diversity (REQ-43) is significantly correlated with class frequency, i.e. whether

minority classes are not merely *smaller* but also *narrower* in visual coverage — a compounding defect that class-rebalancing alone will not fix.

- **REQ-58:** The system shall test for a **Label-Noise–Fairness/Class Confound**: whether the estimated label-issue rate (§4.7) is non-uniform across classes or sensitive groups, identifying sub-populations whose annotations are systematically less reliable.
- **REQ-59:** The system shall test for a **Leakage–Class Confound**: whether cross-split duplicate pairs (REQ-37) concentrate in particular classes, since class-specific leakage inflates the apparent per-class accuracy of exactly those classes.
- **REQ-60:** Each confound finding shall be reported with: the two dimensions involved, the test statistic, the corrected *p*-value, the effect size, the affected group(s), the affected sample count, and a plain-language statement of the downstream consequence (e.g. "The model may learn to associate low illumination with class *benign* rather than learning the lesion morphology itself").
- **REQ-61:** The system shall **explicitly report which confounds were tested and found absent**, so that the absence of a finding is distinguishable from the absence of a test.
- **REQ-62:** Confound findings shall be surfaced in a dedicated **"Cross-Dimensional Findings"** section of the Dataset Health Report (REQ-88), positioned before the per-dimension findings, since an interaction defect is generally more consequential — and less obvious — than any of its constituent single-dimension defects.
- **REQ-63:** Detected confounds shall be admitted as features to the bias descriptor vector (REQ-65) and shall be addressable by the recommendation engine (REQ-80), which shall issue *interaction-aware* remediations (e.g. "do not simply oversample the minority class — its samples are also lower quality and less diverse; collect new, higher-quality samples for it instead").

4.9 Bias Impact Prediction Module (Novel Contribution C2)

4.9.1 Description and Priority

The project's second research contribution. Rather than merely *reporting* bias, the module **predicts the downstream consequence of that bias** — specifically, the expected drop in overall test accuracy and the expected sub-group performance gap that a model trained on this dataset would exhibit. This converts an abstract fairness finding ("gender is 82:18") into an actionable quantity ("expect ≈ 11 pp lower accuracy on the female sub-group").

This module **depends on C1**: its input is a descriptor assembled from *all six* dimensions plus the cross-dimensional confounds of §4.8. A single-axis tool could not construct it, because downstream degradation is a joint function of imbalance, quality, redundancy, diversity, label noise, and their interactions — not of any one of them alone.
Priority: High *(Benefit 9, Penalty 6, Cost 8, Risk 8)*

4.9.2 Stimulus/Response Sequences

1. All prior analyses, including cross-dimensional confound analysis (§4.8), complete → System assembles a **bias descriptor vector** from the computed metrics.

2. System passes the descriptor to the trained Bias Impact Predictor → Predictor returns (a) expected accuracy degradation Δacc relative to a balanced reference, and (b) expected worst-group accuracy gap Δgap, each with a prediction interval.

3. The user requests validation → System reports the predictor's own held-out R^2 / MAE, and permits an *empirical verification run* in which a small surrogate model is genuinely trained on the user's dataset and the prediction is compared against the observed outcome.

4.9.3 Functional Requirements

- **REQ-64:** The system shall construct a **bias descriptor vector** spanning **all six dimensions plus their interactions**, comprising, at minimum: class Imbalance Ratio, class entropy, per-attribute Imbalance Ratios, Cramér's V of class × attribute association, minimum intersectional sub-group proportion, redundancy ratio, cross-split leakage count, low-quality proportion, mean no-reference quality score, diversity entropy, per-class diversity variance, estimated label noise rate, **and the effect sizes of every cross-dimensional confound detected in §4.8**. This vector is constructible **only** because the unified architecture (C1) measures all dimensions over one shared sample index; no single-axis tool could assemble it.
- **REQ-65:** The system shall include a **Bias Impact Predictor**: a supervised regression model (e.g. gradient-boosted trees or a small MLP) mapping the bias descriptor vector to the target pair (Δacc, Δgap).
- **REQ-66:** The Bias Impact Predictor shall be **trained offline** on a corpus of *synthetically perturbed datasets*: from each base dataset (UTKFace, FairFace, CIFAR-10, HAM10000, Chest X-Ray), the system shall generate a large number of controlled variants by deliberately injecting known quantities of class imbalance, demographic skew, duplication, quality degradation, and label noise — **including variants with deliberately co-occurring, interacting defects** (e.g. quality degradation applied preferentially to one demographic sub-group), so that the predictor can learn interaction effects and not merely additive main effects; a fixed surrogate CNN (e.g. ResNet-18) shall be trained on each variant and evaluated on a held-out *balanced* test set. The resulting (descriptor → observed degradation) pairs constitute the training corpus.
- **REQ-67:** The system shall report, for each prediction, a **prediction interval** (e.g. via quantile regression or an ensemble spread), and shall never present the predicted degradation as a point-exact guarantee.
- **REQ-68:** The system shall report the predictor's **out-of-sample validation performance** (MAE, RMSE, R^2, and Spearman rank correlation between predicted and observed degradation) computed on datasets held out from predictor training, and shall include a **leave-one-dataset-out (LODO)** generalisation result to substantiate cross-dataset transfer.
- **REQ-69:** The system shall conduct and report an **ablation over the bias descriptor**, quantifying the loss in predictive accuracy when each dimension's features are withheld. This ablation constitutes the **empirical evidence for contribution C1**: if a predictor restricted to fairness features alone (or quality features alone) performs materially worse than the full six-dimension predictor, then unification is demonstrated to be *necessary*, not

merely *convenient*. The ablation shall specifically report the marginal contribution of the cross-dimensional confound features of §4.8.

- **REQ-70:** The system shall provide **per-feature attribution** for each prediction (e.g. SHAP values over the bias descriptor), so the user can see *which* defect — or which *interaction* of defects — is the dominant contributor to the predicted degradation. This attribution shall directly feed the recommendation engine (§4.11).
- **REQ-71:** The system shall offer an optional **Empirical Verification Mode** which trains the surrogate CNN on the user's actual dataset and reports predicted-versus-observed degradation, thereby providing an on-demand ground-truth check of the predictor.
- **REQ-72:** If the user's dataset lies outside the modality or scale range of the predictor's training corpus, the system shall emit an **extrapolation warning** and shall widen the reported prediction interval accordingly.
- **REQ-73:** The system shall degrade gracefully: if the Bias Impact Predictor is unavailable or refuses to predict (out-of-domain input), all other features (§4.1–§4.8, §4.10–§4.12) shall remain fully functional, and the report shall state that impact prediction was not performed.

4.10 AI Readiness Score Computation

4.10.1 Description and Priority

Condenses all six dimensions, plus their detected interactions, into a **single interpretable 0–100 overall dataset integrity score** with transparent, auditable sub-scores. This feature is the point at which unification (C1) becomes user-visible: it is what places an Imbalance Ratio, a near-duplicate rate, and a BRISQUE score onto **one common, comparable scale**, and thereby makes it possible — for the first time — to answer the practitioner's actual question, *"which of my dataset's problems is the worst one?"*
Priority: High *(Benefit 9, Penalty 8, Cost 3, Risk 4)*

4.10.2 Stimulus/Response Sequences

1. All dimension analyses complete → System normalises each dimension to a 0–100 sub-score and combines them into the composite score.

2. Any dimension sub-score falls below a hard floor → System caps the composite score and marks the dataset **Not Ready**, regardless of strength on other dimensions.

3. A cross-dimensional confound is detected → System applies a confound penalty to *both* participating sub-scores, since the defect belongs to neither alone.

4. User adjusts dimension weights → System recomputes the composite score immediately and shows the delta.

4.10.3 Functional Requirements

- **REQ-74:** The system shall compute six dimension sub-scores, each normalised to `[0,100]` on a **common, documented scale**: **Balance**, **Fairness**, **Quality**, **Uniqueness**

(inverse of redundancy/leakage), **Diversity**, and **Label Integrity**. The normalisation function for each dimension shall be published in the Methodology appendix, since comparability across dimensions is the property that makes the composite meaningful.

- **REQ-75:** The **AI Readiness Score** — the system's **overall dataset integrity score** — shall be the weighted arithmetic mean of the six sub-scores, with default weights Balance 0.20, Fairness 0.20, Quality 0.15, Uniqueness 0.15, Diversity 0.15, Label Integrity 0.15. Weights shall be user-configurable and shall be normalised to sum to 1.
- **REQ-76:** The system shall apply a **confound penalty**: where §4.8 detects a significant cross-dimensional confound with non-trivial effect size, a penalty proportional to the effect size shall be applied to **each** of the two participating sub-scores. A defect arising from the interaction of two dimensions must not be permitted to hide in the gap between them.
- **REQ-77:** The system shall apply a **hard gate**: if any sub-score is below 30, or if any train–test leakage is detected, the composite score shall be capped at 49 and the dataset labelled **Not Ready for Training**, irrespective of the weighted mean.
- **REQ-78:** The system shall map the composite score to a verdict band: **0–39 Critical / Not Usable**, **40–69 Needs Remediation**, **70–84 Acceptable**, **85–100 Production Ready**.
- **REQ-79:** The system shall record and display the exact formula, weights, penalties, and input metric values used to produce the score, so that any score is fully reproducible and auditable from the report alone.
- **REQ-80:** The system shall support **what-if re-scoring**: given a hypothetical remediation (e.g. "remove all duplicates"), the system shall recompute and display the projected score, using the Bias Impact Predictor's attributions where applicable.

4.11 Remediation Recommendation Engine

4.11.1 Description and Priority

Translates findings into a ranked, concrete, cost-aware list of corrective actions. Because all defects are expressed on the common scale of §4.10, the engine can do what no composition of single-axis tools can: **rank remediations across dimensions against one another**, and issue **interaction-aware** advice.
Priority: High *(Benefit 9, Penalty 8, Cost 4, Risk 3)*

4.11.2 Stimulus/Response Sequences

1. Scoring completes → System maps each detected defect to one or more candidate remediations from a rule base.

2. System ranks candidates by projected score improvement (from REQ-80) divided by estimated effort → Ranked list is presented.

3. A cross-dimensional confound is present → System suppresses the naive single-dimension remediation and substitutes an interaction-aware one.

4. User selects a recommendation → System shows the affected sample list and, where applicable, exports a manifest of the affected files (e.g. duplicate file paths to delete).

4.11.3 Functional Requirements

- **REQ-81:** The system shall maintain a **remediation rule base** mapping defect types to actions, including at minimum: oversample/undersample the affected class, apply class-weighted loss, collect additional samples for a named under-represented sub-group, remove duplicate groups, re-split to eliminate leakage, remove or re-capture low-quality samples, apply targeted augmentation (brightness/contrast for lighting bias), prune redundant clusters, and re-annotate the top-ranked suspected mislabels.
- **REQ-82:** Each recommendation shall be issued with: the defect it addresses, the affected sample count, the **projected AI Readiness Score gain**, an **estimated effort** rating (Low / Medium / High), and a one-line rationale.
- **REQ-83:** Recommendations shall be **ranked by projected gain per unit effort across all six dimensions on the common scale**, and the top three shall be surfaced on the Health Dashboard as "Do these first." This cross-dimensional prioritisation is possible only under the unified scoring of §4.10; a practitioner running six separate tools receives six incomparable outputs and no priority order at all.
- **REQ-84:** The system shall issue **interaction-aware remediations** where §4.8 has detected a confound, and shall suppress the naive single-dimension remediation that would otherwise be triggered. For example, where a minority class is found to be *simultaneously* under-represented, lower in image quality, and narrower in diversity, the system shall **not** recommend simple oversampling (which would merely replicate low-quality, low-diversity samples and could worsen the outcome), but shall instead recommend targeted collection of new, high-quality, diverse samples for that class, and shall state this reasoning explicitly.
- **REQ-85:** Recommendations that require *new data collection* shall be stated in specific, actionable terms — naming the exact sub-group and the approximate number of additional samples required to reach a target balance (e.g. "Collect ≈ 1,240 additional images of subjects aged 60+ to bring the age-band Imbalance Ratio below 3.0").
- **REQ-86:** For every recommendation whose target defect is machine-actionable (duplicates, corrupt files, low-quality samples, suspected mislabels), the system shall export a **CSV manifest** of the affected file paths, so the user can act on it programmatically. The system shall not itself delete or modify any user file (per **CON-4**).
- **REQ-87:** The system shall explicitly state when a recommendation carries a **trade-off risk** (e.g. "Undersampling the majority class will reduce total training data by 34 %; prefer class weighting if the dataset is already small").
- **REQ-88:** The system shall avoid contradictory recommendations: the rule base shall detect conflicts (e.g. simultaneously recommending aggressive undersampling and warning of insufficient data) and shall resolve them to a single coherent recommendation before presentation. Conflict resolution is tractable precisely because all recommendations are scored on one scale.

4.12 Dataset Health Report Generation

4.12.1 Description and Priority

Produces the persistent, shareable artifact that is the product's principal deliverable — **one report, covering all six dimensions, replacing the six incomparable outputs of the fragmented toolchain.**
Priority: High *(Benefit 9, Penalty 8, Cost 4, Risk 2)*

4.12.2 Stimulus/Response Sequences

1. Analysis completes → System renders the report in the browser.

2. User clicks Export → System generates the report in the chosen format (PDF / HTML / JSON) and initiates download.

3. User re-opens a past run → System retrieves the stored Analysis Run from SQLite and re-renders the identical report.

4.12.3 Functional Requirements

- **REQ-89:** The Dataset Health Report shall contain the following sections, in order: (1) Executive Summary with the overall AI Readiness Score and verdict; (2) Dataset Overview (fingerprint, sample count, class list, splits, analysis date, sampling rate); (3) Critical Issues (ranked across all dimensions); (4) **Cross-Dimensional Findings** (§4.8); (5) Per-dimension findings with charts and statistics (Balance, Fairness, Quality, Uniqueness, Diversity, Label Integrity); (6) Predicted Bias Impact with prediction interval and feature attributions; (7) Ranked Recommendations; (8) Methodology and Reproducibility appendix (all metric formulae, normalisation functions, thresholds, model versions, library versions, random seed).
- **REQ-90:** The report shall present a **unified summary table** in which every dimension appears with its raw metrics, its normalised 0–100 sub-score, and its verdict — so that all six dimensions can be read and compared at a glance, on one page, on one scale. This table is the primary user-facing artifact of contribution C1.
- **REQ-91:** The report shall be exportable to **PDF**, **standalone HTML** (charts embedded, no external asset dependencies), and **machine-readable JSON** conforming to a published schema.
- **REQ-92:** The JSON export shall be complete and lossless with respect to the report contents, enabling programmatic consumption in CI/CD pipelines and downstream research.
- **REQ-93:** Every Analysis Run shall be persisted with its run identifier, timestamp, dataset fingerprint, configuration (weights, thresholds), and full result set, and shall be retrievable and re-renderable without recomputation.
- **REQ-94:** The system shall support **comparison of two runs** over the same dataset fingerprint, presenting a side-by-side delta of every sub-score and the composite score, so that the effect of a remediation can be demonstrated quantitatively.
- **REQ-95:** The report shall include a **model card / datasheet-style summary block** (after Gebru et al.) suitable for direct inclusion in a publication or a dataset release. Whereas Datasheets for Datasets prescribes a *disclosure template* to be filled in by hand, DIA **populates the quality-relevant fields automatically from measurement**.

- **REQ-96:** All figures in the report shall be reproducible from the persisted Analysis Run alone; no figure shall depend on transient in-memory state.

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5. Other Nonfunctional Requirements

5.1 Performance Requirements

Performance is stated against a **reference configuration**: 8-core CPU, 16 GB RAM, no GPU; and against an **accelerated configuration**: the same host with a single NVIDIA GPU ≥ 8 GB VRAM.

- **NFR-P1:** The system shall complete a full analysis of a **10,000-image dataset** (≈ 224×224 px) within **10 minutes** on the reference configuration and within **3 minutes** on the accelerated configuration.
  Rationale: the tool must be materially faster than the training run it is meant to protect, or it will not be used.
- **NFR-P2:** The system shall complete ingestion and schema inference for a 10,000-image dataset within **30 seconds**.
- **NFR-P3:** Duplicate detection shall scale as **O(N log N)** or better in the number of samples; an $O(N^2)$ exhaustive pairwise comparison is explicitly prohibited (see REQ-35).
- **NFR-P4:** The dashboard shall render the Health Dashboard screen within **2 seconds** of the analysis result being available.
- **NFR-P5:** Peak resident memory shall not exceed **8 GB** for datasets up to 100,000 images; the system shall achieve this through batched, streaming processing and by persisting embeddings to disk rather than holding them all in RAM.
- **NFR-P6:** Analysis progress shall be reported to the client at intervals of **≤ 2 seconds**, so that a long-running analysis never appears frozen.
- **NFR-P7:** The system shall support **cancellation** of a running analysis, releasing all resources within 5 seconds of the request.
- **NFR-P8:** Re-running an analysis on an unchanged dataset (identical fingerprint) with an unchanged configuration shall complete in **≤ 5 seconds** by serving cached embeddings and per-sample metrics.
- **NFR-P9:** Bias Impact Prediction (inference, not the offline training of REQ-55) shall complete in **≤ 1 second** once the bias descriptor vector is available.

5.2 Safety Requirements

The system performs analysis only and controls no physical process; the safety surface is therefore confined to *decision harm* — harm arising from a user acting on an erroneous or over-trusted output.

- **NFR-S1:** The system **shall never modify, move, rename, or delete any file within the user's dataset directory.** All dataset access shall be strictly read-only, enforced at the file-open call. This prevents irrecoverable loss of research data.
- **NFR-S2:** Recommendations that would result in data loss (e.g. "delete these 412 duplicates") shall be issued as an **exported manifest for the user to act upon**, never as an automatic action.
- **NFR-S3:** Inferred demographic attributes shall carry a mandatory, non-suppressible disclaimer (REQ-15). Presenting model-inferred race/gender/age as fact could cause the user to draw and publish false conclusions about individuals or populations, and could itself perpetuate representational harm.
- **NFR-S4:** The predicted bias impact (Section 4.8) shall always be presented **with its prediction interval and its validation error**, and shall be labelled an *estimate*. The system shall not present a predicted accuracy degradation as a guaranteed outcome.
- **NFR-S5:** The system shall emit an explicit **out-of-domain warning** (REQ-60) when analysing a dataset materially unlike those on which the predictor was trained, so that the user does not over-trust an extrapolated prediction — particularly in sensitive domains such as medical imaging (HAM10000, Chest X-Ray).
- **NFR-S6:** The system shall not be represented, in its UI or documentation, as a certification of fairness or regulatory compliance. The report shall carry a statement that it is a diagnostic aid and does not constitute a legal or regulatory fairness audit.
- **NFR-S7:** The system shall fail safe: any analysis dimension that errors shall be reported as **"Not Assessed"** in the report, and shall **not** contribute a default or optimistic value to the AI Readiness Score. A score computed from incomplete evidence shall be explicitly marked as partial.

5.3 Security Requirements

- **NFR-SEC1:** The system shall not transmit dataset content, image data, extracted embeddings, or derived metrics to any external host under any circumstance. All processing is local (**CON-5**).
- **NFR-SEC2:** The web server shall bind to `127.0.0.1` by default. Binding to a network-visible interface shall require an explicit configuration change and shall, in that case, require **TLS 1.2+** (**CI-4**).
- **NFR-SEC3:** Where the system is deployed multi-user, it shall enforce authentication (username/password with passwords stored as salted **bcrypt** or **Argon2id** hashes) and shall scope every Analysis Run to its owning user. Single-user local deployment (the default) may operate without authentication.
- **NFR-SEC4:** All user-supplied paths shall be canonicalised and validated to prevent **path traversal**; the system shall refuse to read outside the designated dataset root.
- **NFR-SEC5:** ZIP extraction shall be protected against **zip-slip** (entries with `..` or absolute paths) and **zip-bomb** (declared uncompressed size and entry count exceeding configurable ceilings) attacks. Malformed archives shall be rejected, not partially extracted.

- **NFR-SEC6:** Uploaded files shall be validated by **magic-number inspection**, not by file extension alone, and shall never be executed.
- **NFR-SEC7:** All REST inputs shall be schema-validated (Pydantic); malformed requests shall be rejected with HTTP 422 and shall not reach the analysis engine.
- **NFR-SEC8:** The system shall emit **no telemetry and no analytics**. This shall be stated in the README and shall be verifiable from the source.
- **NFR-SEC9:** Pre-trained model weights fetched at first run shall be verified against a **published checksum** before being loaded, to prevent loading of a tampered artifact.
- **NFR-SEC10:** Because analysed datasets may contain personally identifiable imagery (faces), the system shall store only derived metrics and low-resolution thumbnails of *exemplar* samples in its database — never full-resolution copies of user images.

5.4 Software Quality Attributes

Stated in relative order of preference: **Correctness** > **Reliability** > **Usability** > **Maintainability** > **Performance** > **Portability**. Where a trade-off is forced, a correct-but-slower result is always preferred to a fast-but-wrong one, because the entire value proposition of the product is trustworthy measurement.

Attribute	Requirement
Correctness	**NFR-Q1:** Every statistical metric shall be unit-tested against an analytically derived expected value on a synthetic fixture dataset with known ground truth (e.g. a dataset constructed with an exact 4:1 imbalance shall report IR = 4.00 ± 0.01).
Reliability	**NFR-Q2:** A corrupt, unreadable, or anomalous individual sample shall never abort a run. The system shall complete analysis of ≥ 99 % of well-formed samples in the presence of up to 5 % corrupt files. **NFR-Q3:** Mean time between analysis-aborting failures shall exceed 100 consecutive runs on the reference datasets.
Reproducibility	**NFR-Q4:** Given the same dataset fingerprint, the same configuration, and the same random seed, two runs shall produce **bit-identical** AI Readiness Scores and metric values. All stochastic components shall accept and honour a seed.
Usability	**NFR-Q5:** A user with basic ML familiarity shall be able to run a first analysis and interpret the resulting score without consulting documentation beyond the quick-start. **NFR-Q6:** Every metric shown in the UI shall have an inline tooltip giving its definition, range, and interpretation.
Testability	**NFR-Q7:** Line coverage of the analysis engine (§4.2–§4.10) shall be ≥ 80 %. Each of REQ-1 … REQ-96 shall be traceable to at least one automated test case. The cross-dimensional confound tests (§4.8)

Attribute	Requirement
	shall additionally be validated against synthetic fixtures in which a confound of known effect size has been deliberately injected.
Maintainability	**NFR-Q8:** Each analysis dimension shall be an independently replaceable module conforming to a common `Analyzer` interface (`run(dataset) -> DimensionResult`), **and shall write its per-sample outputs into the shared unified sample record (REQ-53) rather than into a private store.** Adding a new dimension shall therefore require no modification to the orchestrator, to the confound engine, or to the scorer — the new dimension becomes automatically available to cross-dimensional analysis, to the bias descriptor, and to the composite score. Unification must be a property of the architecture, not a one-off integration. **NFR-Q9:** All thresholds and weights shall be externalised to a YAML configuration file — no magic numbers in source.
Portability	**NFR-Q10:** The system shall run without source modification on Linux, Windows, and macOS, and shall run identically inside the supplied Docker image.
Interoperability	**NFR-Q11:** The JSON report schema shall be published and versioned, so that third-party tooling and CI pipelines can consume DIA output.
Robustness	**NFR-Q12:** The system shall handle degenerate inputs without crashing: single-class datasets (rejected with a clear message), single-sample classes, images of 1×1 px, and datasets in which every sample is a duplicate of every other.
Extensibility	**NFR-Q13:** Adding support for a new sensitive attribute shall require only registering a new attribute inferrer and a configuration entry — no change to the fairness analysis or scoring code.

5.5 Business Rules

- **BR-1:** The AI Readiness Score shall **always** be accompanied by its constituent sub-scores. The composite score shall never be displayed, exported, or transmitted in isolation, because a single number without its decomposition invites exactly the uncritical trust the product exists to prevent.
- **BR-2:** Ground-truth attributes, where supplied, **always** override model-inferred attributes (REQ-13). Inference is a fallback, never a preference.
- **BR-3:** Any dataset exhibiting **train–test leakage** shall be marked **Not Ready for Training**, regardless of how strong its other dimensions are (REQ-77). Leakage invalidates the evaluation itself, so no other virtue can compensate.
- **BR-4:** A dimension that could not be assessed shall be reported as **Not Assessed** and shall be excluded from the weighted mean with the weights renormalised — it shall never be silently imputed as a passing value (**NFR-S7**).

- **BR-5:** The system shall never modify the user's dataset. Only the *user* may act on a recommendation (**NFR-S1**, **NFR-S2**).
- **BR-6:** Only an authenticated user (in multi-user deployments) may view or export the Analysis Runs they own; runs are not shared by default.
- **BR-7:** Only an administrator may alter the global default weights and thresholds in the configuration file. An ordinary user may override weights **for their own run only**, and any such override shall be recorded in the report so that a non-default score is never mistaken for a default one.
- **BR-8:** Every published metric shall cite its source formula in the Methodology appendix. No metric may appear in the report without a defined, citable derivation.
- **BR-9:** No dimension may be reported in isolation. Every analysis run shall assess **all six dimensions** (or explicitly mark a dimension **Not Assessed** per **BR-4**) and shall present them together on the common scale. A user may not configure the system to emit, for example, a fairness-only report, because doing so would reproduce exactly the single-axis fragmentation the product exists to eliminate — and, worse, would licence a naive single-dimension remediation that a confound finding might have contraindicated (REQ-84).
- **BR-10:** Where a cross-dimensional confound is detected, the interaction-aware recommendation shall **supersede**, not merely accompany, the naive single-dimension recommendation for the same defect. Presenting both would return to the user precisely the ambiguity that unification is meant to resolve.

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6. Other Requirements

Database requirements

- **OR-1:** The system shall persist, per Analysis Run: `run_id` (UUID), `dataset_fingerprint`, `created_at`, `config_json`, `sample_count`, `class_manifest`, a per-sample metric table (`sample_id`, `path_hash`, `class`, inferred attributes + confidences, quality metrics, embedding reference, duplicate group, label-issue flag), per-dimension aggregate results, the score record, and the recommendation list.
- **OR-2:** Embeddings shall be stored outside the relational store as memory-mapped NumPy arrays (`.npy`), referenced by offset, to keep the SQLite database compact.
- **OR-3:** The schema shall be versioned and migrated forward with a migration tool (e.g. Alembic); an older database shall never be silently misread by a newer binary.
- **OR-4:** Runs shall be retained indefinitely by default, with an administrator-configurable retention policy and an explicit user-initiated delete.

Legal and licensing requirements

- **OR-5:** All bundled dependencies shall carry permissive licences (**CON-3**); a `THIRD_PARTY_LICENSES` file shall enumerate every dependency and its licence.
- **OR-6:** The system shall not redistribute any benchmark dataset. Evaluation datasets (UTKFace, FairFace, CelebA, HAM10000, Chest X-Ray, CIFAR, COCO/ImageNet subsets) shall be obtained by the user from their original sources (Kaggle, GitHub, IEEE DataPort) under their own licence terms; the repository shall ship only download scripts and citations.
- **OR-7:** Because the system processes facial imagery, the documentation shall state the user's responsibility to hold a lawful basis for processing such data under applicable data-protection law (e.g. GDPR Art. 9 for biometric data). The system's fully-local, no-transmission design (**NFR-SEC1**) is a deliberate measure in support of this.

Reuse objectives

- **OR-8:** The analysis engine shall be importable as a standalone Python package (`import dia`) independently of the web layer, so that it can be embedded in a user's own training pipeline or CI job.
- **OR-9:** The synthetic-perturbation harness built for REQ-66 (controlled injection of imbalance, skew, duplication, quality degradation, and label noise) shall be released as a reusable, independently useful benchmark-generation utility, and shall be a secondary contribution of the associated publication.

Internationalisation

- **OR-10:** All user-facing strings shall be externalised to resource files to permit future localisation. Version 1.0 ships English (`en-US`) only.

Appendix A: Glossary

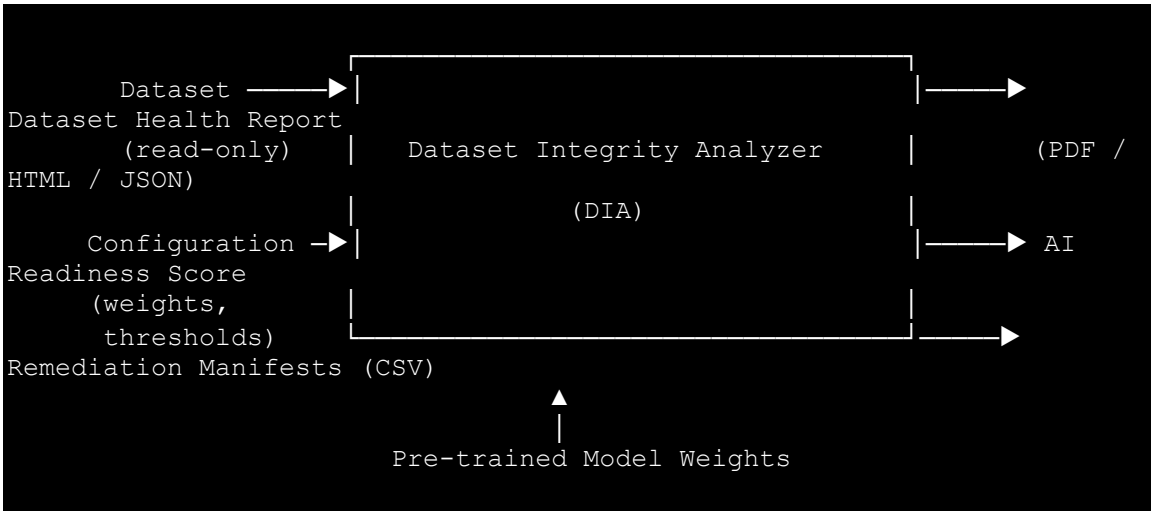
Term	Definition
AI Readiness Score	A composite 0–100 metric produced by this system, summarising a dataset's fitness for model training across six weighted dimensions.

Term	Definition
Bias Descriptor Vector	The fixed-length numeric vector of dataset-level bias and quality statistics that is the input to the Bias Impact Predictor.
Bias Impact Prediction	The estimation of the downstream degradation in model accuracy and sub-group fairness attributable to a dataset's measured defects. The novel contribution of this project.
BRISQUE	Blind/Referenceless Image Spatial Quality Evaluator — a no-reference perceptual image-quality metric.
Class Imbalance	Unequal representation of target classes within a dataset.
Confident Learning	A framework for identifying probable label errors from a model's out-of-fold predicted probabilities.
Coverage Gap	An intersectional sub-group with insufficient or zero representation in the dataset.
Cross-Dimensional Confound	A dataset defect arising from the *interaction* of two quality dimensions (e.g. image quality correlated with a demographic group). Structurally invisible to any tool that measures only one dimension.
Overall Dataset Integrity Score	Synonym for the AI Readiness Score: the single 0–100 composite that places all six dimensions on one comparable scale.
Single-Axis Tool	An existing tool or method that assesses exactly one dimension of dataset quality (e.g. AIF360 for fairness, imagededup for duplication). The fragmented baseline this work unifies.
Unified Dataset Health Framework	A system in which all dimensions of dataset quality are measured over a single shared sample index and embedding store, yielding one comparable scale, one composite score, and the ability to detect cross-dimensional confounds.
Cramér's V	A measure of association between two nominal variables, derived from χ^2, bounded in [0,1].
Demographic Parity Difference	The absolute difference in representation (or positive-outcome rate) between demographic groups.
Disparate Impact Ratio	The ratio of the rate for the unprivileged group to that of the privileged group; values far from 1.0 indicate disparity.
DIA	Dataset Integrity Analyzer — the system specified by this document.
Embedding	A fixed-length vector representation of an image produced by a CNN's penultimate layer.
Imbalance Ratio (IR)	n(majority class) / n(minority class).

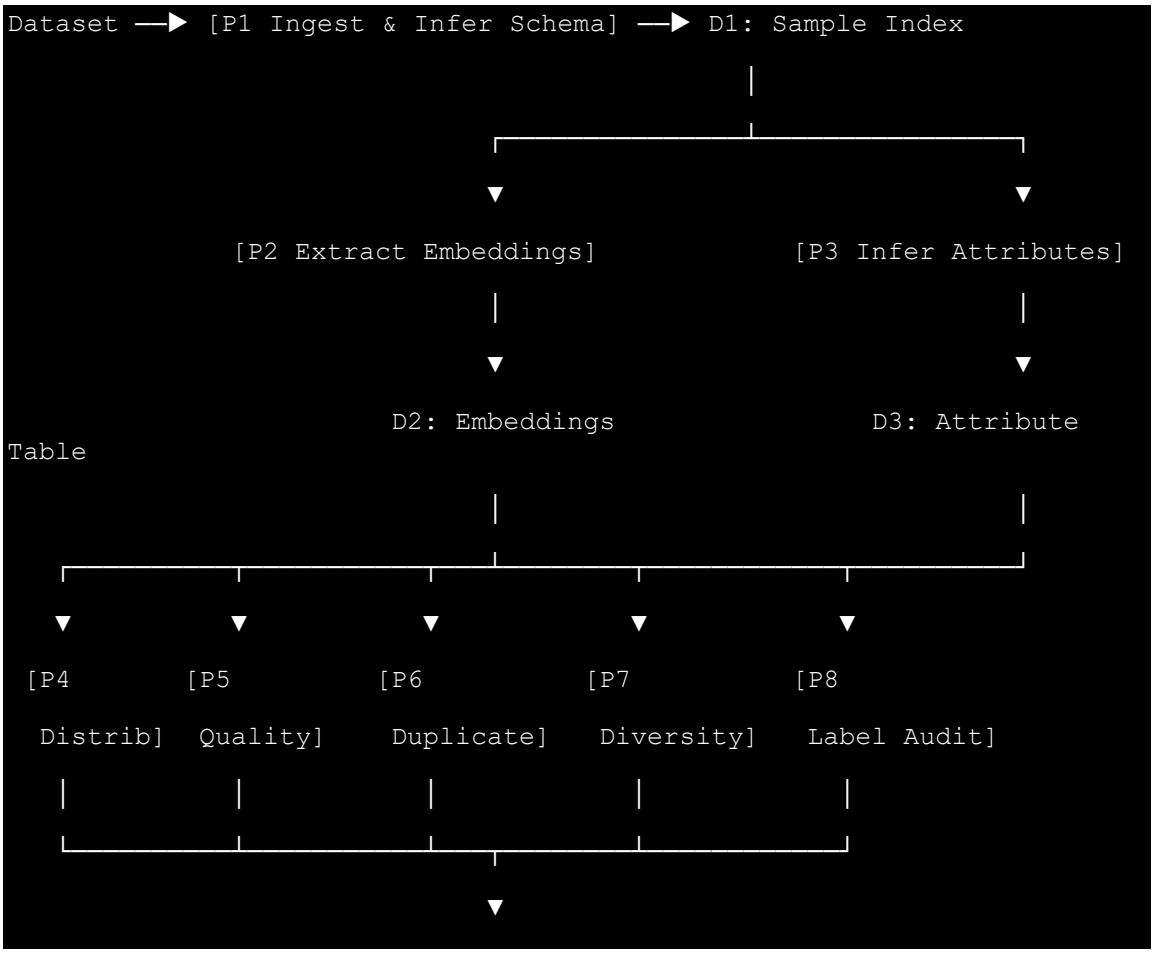
Term	Definition
Jensen–Shannon Divergence	A bounded, symmetric measure of the difference between two probability distributions; used here to detect split skew.
LODO	Leave-One-Dataset-Out — a cross-validation protocol in which an entire dataset is held out to test cross-dataset generalisation.
Near-Duplicate	A pair of samples that are not byte-identical but are perceptually or semantically indistinguishable.
No-Reference IQA	Image quality assessment performed without access to a pristine reference image.
Perceptual Hash (pHash)	A hash whose Hamming distance is small for visually similar images.
Quality Confound	A statistically significant association between an image-quality defect and a class or demographic group — a covert bias source.
Redundancy Ratio	The proportion of samples in a dataset that are exact or near-duplicates of another sample.
Sensitive / Protected Attribute	An attribute (gender, age, skin tone, etc.) across which fairness is evaluated.
SHAP	SHapley Additive exPlanations — a game-theoretic feature-attribution method, used here to attribute a predicted degradation to individual bias descriptors.
Split Skew	A material difference in the distribution of classes or attributes between the train, validation, and test splits.
Surrogate Model	The lightweight CNN (e.g. ResNet-18) trained on perturbed dataset variants to generate ground-truth degradation labels for the Bias Impact Predictor.
Train–Test Leakage	The presence of duplicate or near-duplicate samples across the train and test splits, which inflates measured accuracy.
UMAP	Uniform Manifold Approximation and Projection — a dimensionality-reduction technique used here to visualise the embedding space.

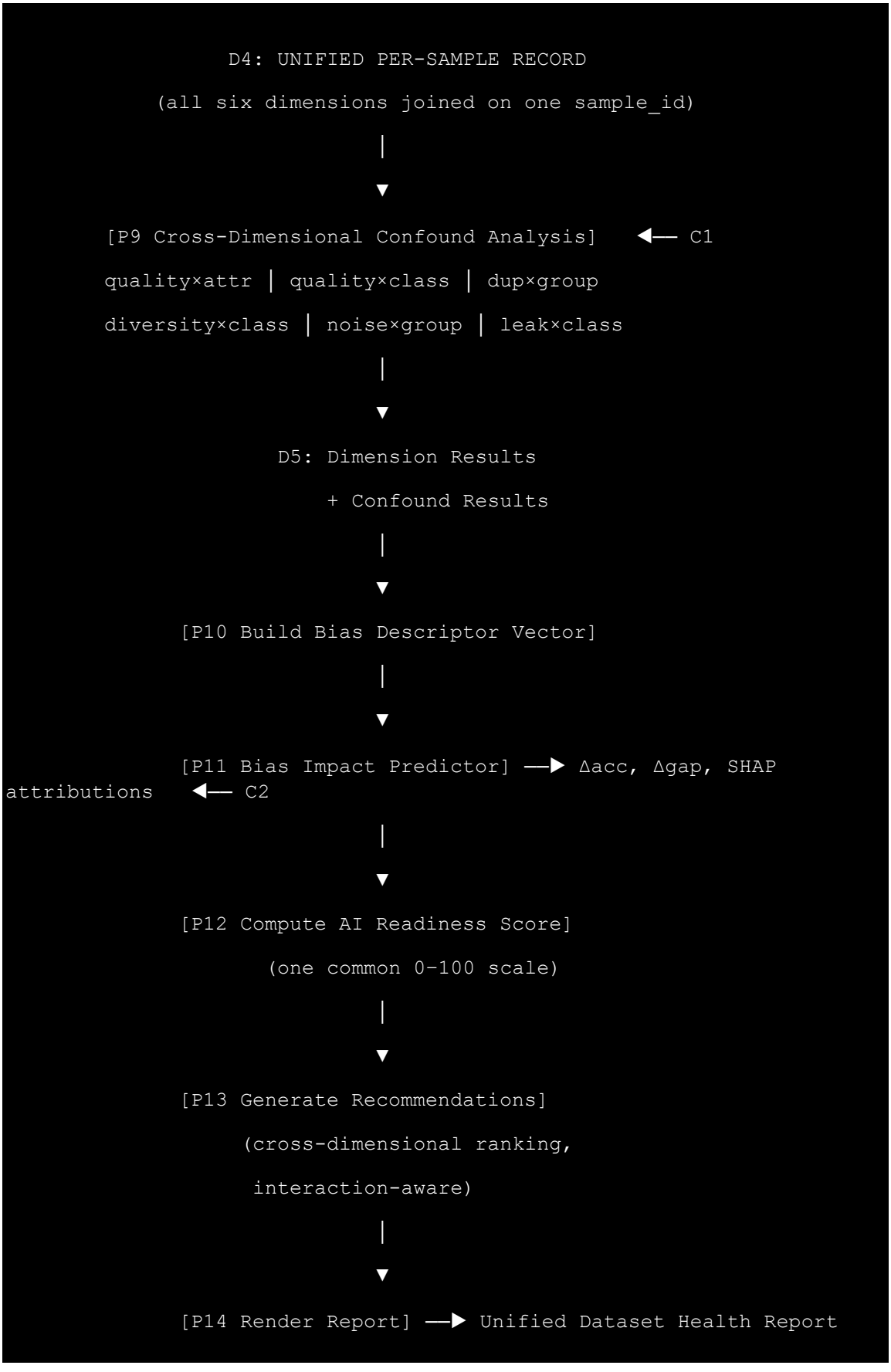
Appendix B: Analysis Models

B.1 Level-0 Data Flow Diagram (Context)



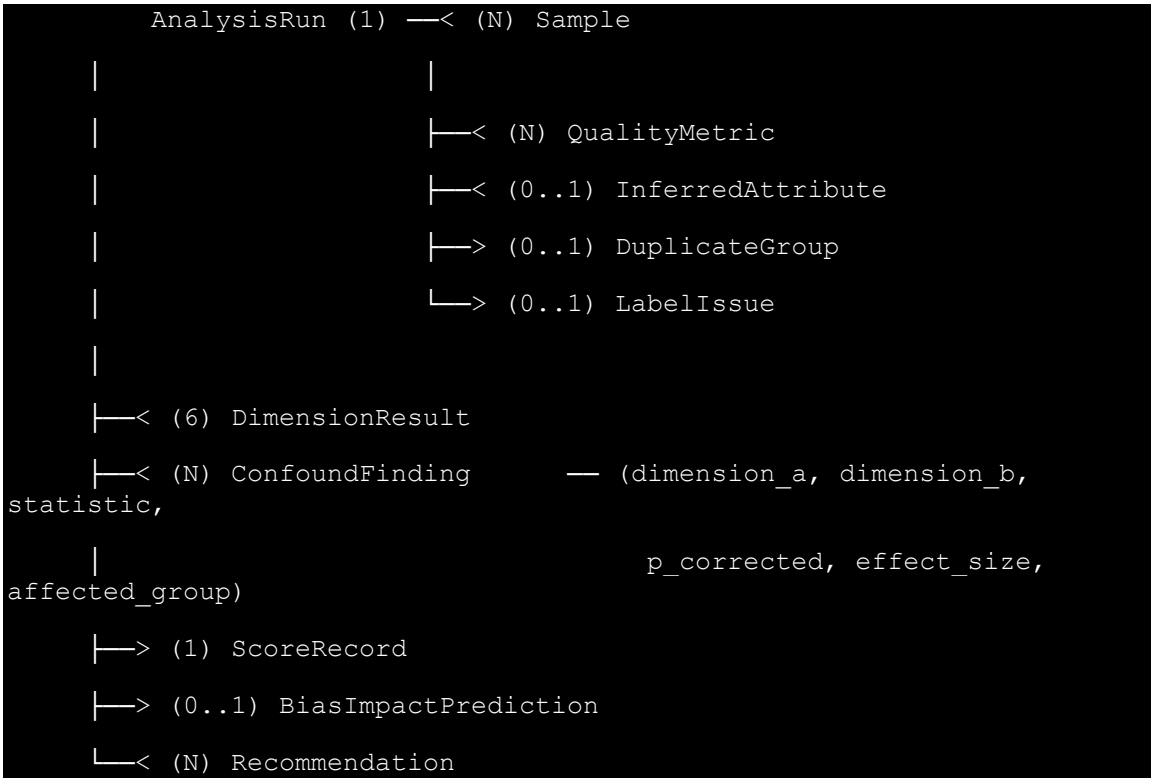
B.2 Level-1 Data Flow Diagram




D4: UNIFIED PER-SAMPLE RECORD
(all six dimensions joined on one sample_id)
[P9 Cross-Dimensional Confound Analysis] ◀── C1
quality×attr | quality×class | dup×group
diversity×class | noise×group | leak×class
D5: Dimension Results
+ Confound Results
[P10 Build Bias Descriptor Vector]
[P11 Bias Impact Predictor] ──▶ Δacc, Δgap, SHAP attributions ◀── C2
[P12 Compute AI Readiness Score]
(one common 0–100 scale)
[P13 Generate Recommendations]
(cross-dimensional ranking,
interaction-aware)
[P14 Render Report] ──▶ Unified Dataset Health Report

The convergence at D4 is the architectural heart of contribution C1. Every dimension writes into one joined per-sample record. P9, P10, P12, and P13 all read from it, and none of them could exist without it: a toolchain of six separate programs produces six disjoint outputs that never meet at a D4, and therefore can support no confound analysis (P9), no bias descriptor (P10), no common scale (P12), and no cross-dimensional prioritisation (P13).

B.3 Principal Entity–Relationship Model



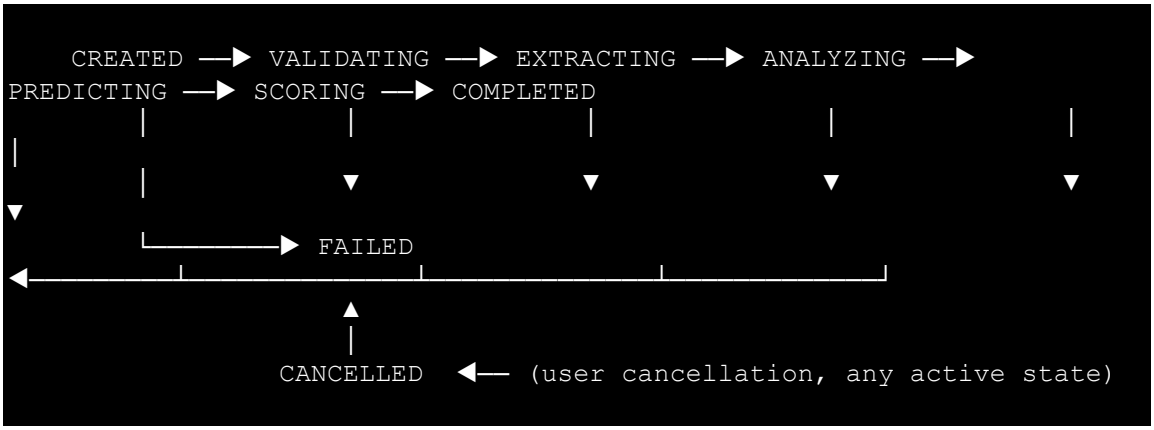
Note that `Sample` is a **single entity carrying the per-sample outputs of all six dimensions** — quality metrics, inferred attributes, duplicate-group membership, cluster membership, and label-issue flag all hang off one `sample_id`. This is the data-model expression of C1, and it is what makes `ConfoundFinding` computable by a simple join.

B.4 Use-Case Summary

ID	Actor	Use Case
UC-1	ML Practitioner	Analyse a new dataset and obtain an AI Readiness Score
UC-2	ML Researcher	Inspect per-dimension findings and exemplar offending samples

ID	Actor	Use Case
UC-2b	ML Researcher	Inspect cross-dimensional confound findings (e.g. quality × demographic)
UC-3	ML Researcher	Run Empirical Verification Mode to validate the predicted bias impact
UC-4	ML Practitioner	Export remediation manifests and act on them
UC-5	ML Practitioner	Re-analyse a remediated dataset and compare runs
UC-6	Auditor	Export and read the Dataset Health Report without running an analysis
UC-7	Student	View a metric definition tooltip and interpret a sub-score
UC-8	Administrator	Configure global weights, thresholds, and resource limits
UC-9	ML Practitioner	Perform a what-if re-score under a hypothetical remediation

B.5 Analysis State Model (Analysis Run lifecycle)



Appendix C: To Be Determined List

TBD	Description	Owner	Target Resolution
TBD-1	Final skin-tone taxonomy: full 6-level Fitzpatrick vs. a coarse 3-level proxy. Depends on the achievable accuracy of the	Developer	Before implementation of Section 4.2

TBD	Description	Owner	Target Resolution
	inference model and on the ethical defensibility of the finer taxonomy. Affects REQ-10.		
TBD-2	Final default weight vector for the six dimensions (REQ-75), and the functional form of the confound penalty (REQ-76). Defaults are provisional and will be calibrated once the Bias Impact Predictor's feature attributions indicate the empirical influence of each dimension on downstream degradation.	Developer	After Section 4.8 validation
TBD-3	Selection of the surrogate architecture for REQ-66 (ResNet-18 vs. MobileNetV3) and the number of perturbed variants per base dataset, both bounded by the available compute budget.	Developer	Before predictor training
TBD-4	Whether Empirical Verification Mode (REQ-71) ships in Version 1.0 or is deferred to 1.1, contingent on training-time feasibility on the reference hardware.	Developer / Faculty	Mid-project review
TBD-5	Final selection of the no-reference IQA metric (BRISQUE vs. NIQE vs. a learned alternative), pending a comparative evaluation on the reference datasets.	Developer	Before implementation of Section 4.4
TBD-6	Front-end stack decision: full React SPA vs. Streamlit (lower effort, reduced UI control). Affects Sections 3.1 and CON-1.	Developer	Before implementation of Section 3.1
TBD-7	Definitive threshold values for all severity bands (REQ-22, REQ-27, REQ-33, REQ-34, REQ-44, REQ-76), to be empirically calibrated on the reference datasets rather than fixed a priori.	Developer	During integration testing